

Spectral Theory of Unsigned and Signed Graphs Applications to Graph Clustering

Some Slides

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Contents

1	Graphs and Graph Laplacians	5
1.1	Directed, Undirected, and Weighted Graphs	5
1.2	Laplacian Matrices of Graphs	22
2	Spectral Graph Drawing	37
2.1	Graph Drawing and Energy Minimization	37
2.2	Examples of Graph Drawings	45
3	Graph Clustering	55
3.1	Graph Clustering Using Normalized Cuts .	55
3.2	Special Case: 2-Way Clustering Using Normalized Cuts	60
3.3	K -Way Clustering Using Normalized Cuts	79
3.4	Discrete Solution Close to a Continuous Approximation	116
4	Signed Graphs	159
4.1	Signed Graphs and Signed Laplacians . . .	159
4.2	Signed Normalized Cuts	169

4.3	Balanced Graphs	176
4.4	K -Way Clustering of Signed Graphs . . .	187
4.5	Signed Graph Drawing	191
5	Graph Clustering Using Ratio Cuts	201
A	Rayleigh Ratios and the Courant-Fischer Theorem	211
	Bibliography	217

Chapter 1

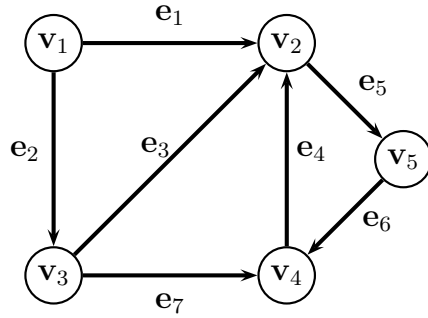
Graphs and Graph Laplacians

1.1 Directed Graphs, Undirected Graphs, Incidence Matrices, Adjacency Matrices, Weighted Graphs

Definition 1.1. A *directed graph* is a pair $G = (V, E)$, where $V = \{v_1, \dots, v_m\}$ is a set of *nodes* or *vertices*, and $E \subseteq V \times V$ is a set of ordered pairs of distinct nodes (that is, pairs $(u, v) \in V \times V$ with $u \neq v$), called *edges*. Given any edge $e = (u, v)$, we let $s(e) = u$ be the *source* of e and $t(e) = v$ be the *target* of e .

Remark: Since an edge is a pair (u, v) with $u \neq v$, self-loops are not allowed.

Also, there is at most one edge from a node u to a node v . Such graphs are sometimes called *simple graphs*.

Figure 1.1: Graph G_1 .

For every node $v \in V$, the *degree* $d(v)$ of v is the number of edges leaving or entering v :

$$d(v) = |\{u \in V \mid (v, u) \in E \text{ or } (u, v) \in E\}|.$$

We abbreviate $d(v_i)$ as d_i . The *degree matrix* $D(G)$, is the diagonal matrix

$$D(G) = \text{diag}(d_1, \dots, d_m).$$

For example, for graph G_1 , we have

$$D(G_1) = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

Unless confusion arises, we write D instead of $D(G)$.

Definition 1.2. Given a directed graph $G = (V, E)$, for any two nodes $u, v \in V$, a *path from u to v* is a sequence of nodes $(v_0, v_1, \dots, v_k)$ such that $v_0 = u$, $v_k = v$, and (v_i, v_{i+1}) is an edge in E for all i with $0 \leq i \leq k - 1$. The integer k is the *length* of the path. A path is *closed* if $u = v$. The graph G is *strongly connected* if for any two distinct node $u, v \in V$, there is a path from u to v and there is a path from v to u .

Remark: The terminology *walk* is often used instead of *path*, the word path being reserved to the case where the nodes v_i are all distinct, except that $v_0 = v_k$ when the path is closed.

The binary relation on $V \times V$ defined so that u and v are related iff there is a path from u to v and there is a path from v to u is an equivalence relation whose equivalence classes are called the *strongly connected components* of G .

Definition 1.3. Given a directed graph $G = (V, E)$, with $V = \{v_1, \dots, v_m\}$, if $E = \{e_1, \dots, e_n\}$, then the *incidence matrix* $B(G)$ of G is the $m \times n$ matrix whose entries b_{ij} are given by

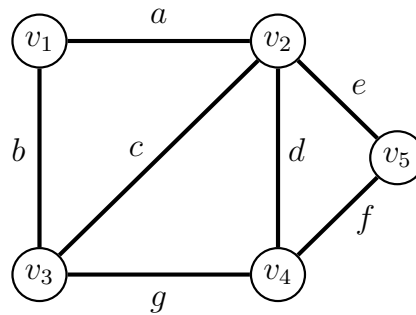
$$b_{ij} = \begin{cases} +1 & \text{if } s(e_j) = v_i \\ -1 & \text{if } t(e_j) = v_i \\ 0 & \text{otherwise.} \end{cases}$$

Here is the incidence matrix of the graph G_1 :

$$B = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & -1 & -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{pmatrix}.$$

Again, unless confusion arises, we write B instead of $B(G)$.

Remark: Some authors adopt the opposite convention of sign in defining the incidence matrix, which means that their incidence matrix is $-B$.

Figure 1.2: The undirected graph G_2 .

Undirected graphs are obtained from directed graphs by forgetting the orientation of the edges.

Definition 1.4. A *graph* (or *undirected graph*) is a pair $G = (V, E)$, where $V = \{v_1, \dots, v_m\}$ is a set of *nodes* or *vertices*, and E is a set of two-element subsets of V (that is, subsets $\{u, v\}$, with $u, v \in V$ and $u \neq v$), called *edges*.

Remark: Since an edge is a set $\{u, v\}$, we have $u \neq v$, so self-loops are not allowed. Also, for every set of nodes $\{u, v\}$, there is at most one edge between u and v .

As in the case of directed graphs, such graphs are sometimes called *simple graphs*.

For every node $v \in V$, the *degree* $d(v)$ of v is the number of edges incident to v :

$$d(v) = |\{u \in V \mid \{u, v\} \in E\}|.$$

The degree matrix D is defined as before.

Definition 1.5. Given a (undirected) graph $G = (V, E)$, for any two nodes $u, v \in V$, a *path from u to v* is a sequence of nodes $(v_0, v_1, \dots, v_k)$ such that $v_0 = u$, $v_k = v$, and $\{v_i, v_{i+1}\}$ is an edge in E for all i with $0 \leq i \leq k-1$. The integer k is the *length* of the path. A path is *closed* if $u = v$. The graph G is *connected* if for any two distinct nodes $u, v \in V$, there is a path from u to v .

Remark: The terminology *walk* or *chain* is often used instead of *path*, the word *path* being reserved to the case where the nodes v_i are all distinct, except that $v_0 = v_k$ when the path is closed.

The binary relation on $V \times V$ defined so that u and v are related iff there is a path from u to v is an equivalence relation whose equivalence classes are called the *connected components* of G .

The notion of incidence matrix for an undirected graph is not as useful as in the case of directed graphs

Definition 1.6. Given a graph $G = (V, E)$, with $V = \{v_1, \dots, v_m\}$, if $E = \{e_1, \dots, e_n\}$, then the *incidence matrix* $B(G)$ of G is the $m \times n$ matrix whose entries b_{ij} are given by

$$b_{ij} = \begin{cases} +1 & \text{if } e_j = \{v_i, v_k\} \text{ for some } k \\ 0 & \text{otherwise.} \end{cases}$$

Unlike the case of directed graphs, the entries in the incidence matrix of a graph (undirected) are nonnegative. We usually write B instead of $B(G)$.

The notion of adjacency matrix is basically the same for directed or undirected graphs.

Definition 1.7. Given a directed or undirected graph $G = (V, E)$, with $V = \{v_1, \dots, v_m\}$, the *adjacency matrix* $A(G)$ of G is the symmetric $m \times m$ matrix (a_{ij}) such that

(1) If G is directed, then

$$a_{ij} = \begin{cases} 1 & \text{if there is some edge } (v_i, v_j) \in E \\ & \text{or some edge } (v_j, v_i) \in E \\ 0 & \text{otherwise.} \end{cases}$$

(2) Else if G is undirected, then

$$a_{ij} = \begin{cases} 1 & \text{if there is some edge } \{v_i, v_j\} \in E \\ 0 & \text{otherwise.} \end{cases}$$

As usual, unless confusion arises, we write A instead of $A(G)$.

Here is the adjacency matrix of both graphs G_1 and G_2 :

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}.$$

If $G = (V, E)$ is a directed or an undirected graph, given a node $u \in V$, any node $v \in V$ such that there is an edge (u, v) in the directed case or $\{u, v\}$ in the undirected case is called *adjacent to v* , and we often use the notation

$$u \sim v.$$

Observe that the binary relation $\sim$ is symmetric when G is an undirected graph, but in general it is not symmetric when G is a directed graph.

If $G = (V, E)$ is an undirected graph, the adjacency matrix A of G can be viewed as a linear map from $\mathbb{R}^V$ to $\mathbb{R}^V$, such that for all $x \in \mathbb{R}^m$, we have

$$(Ax)_i = \sum_{j \sim i} x_j;$$

that is, the value of Ax at v_i is the sum of the values of x at the nodes v_j adjacent to v_i .

The adjacency matrix can be viewed as a *diffusion operator*.

This observation yields a geometric interpretation of what it means for a vector $x \in \mathbb{R}^m$ to be an eigenvector of A associated with some eigenvalue λ ; we must have

$$\lambda x_i = \sum_{j \sim i} x_j, \quad i = 1, \dots, m,$$

which means that the the sum of the values of x assigned to the nodes v_j adjacent to v_i is equal to λ times the value of x at v_i .

Definition 1.8. Given any undirected graph $G = (V, E)$, an *orientation* of G is a function $\sigma: E \rightarrow V \times V$ assigning a source and a target to every edge in E , which means that for every edge $\{u, v\} \in E$, either $\sigma(\{u, v\}) = (u, v)$ or $\sigma(\{u, v\}) = (v, u)$. The *oriented graph* G^σ obtained from G by applying the orientation σ is the directed graph $G^\sigma = (V, E^\sigma)$, with $E^\sigma = \sigma(E)$.

Proposition 1.1. *Let $G = (V, E)$ be any undirected graph with m vertices, n edges, and c connected components. For any orientation σ of G , if B is the incidence matrix of the oriented graph G^σ , then $c = \dim(\text{Ker}(B^\top))$, and B has rank $m - c$. Furthermore, the nullspace of $B^\top$ has a basis consisting of indicator vectors of the connected components of G ; that is, vectors $(z_1, \dots, z_m)$ such that $z_j = 1$ iff v_j is in the i th component K_i of G , and $z_j = 0$ otherwise.*

Following common practice, we denote by $\mathbf{1}$ the (column) vector whose components are all equal to 1. Observe that

$$B^\top \mathbf{1} = 0.$$

According to Proposition 1.1, the graph G is connected iff B has rank $m - 1$ iff the nullspace of $B^\top$ is the one-dimensional space spanned by $\mathbf{1}$.

In many applications, the notion of graph needs to be generalized to capture the intuitive idea that two nodes u and v are linked with a degree of certainty (or strength).

Thus, we assign a nonnegative weight w_{ij} to an edge $\{v_i, v_j\}$; the smaller w_{ij} is, the weaker is the link (or similarity) between v_i and v_j , and the greater w_{ij} is, the stronger is the link (or similarity) between v_i and v_j .

Definition 1.9. A *weighted graph* is a pair $G = (V, W)$, where $V = \{v_1, \dots, v_m\}$ is a set of *nodes* or *vertices*, and W is a symmetric matrix called the *weight matrix*, such that $w_{ij} \geq 0$ for all $i, j \in \{1, \dots, m\}$, and $w_{ii} = 0$ for $i = 1, \dots, m$. We say that a set $\{v_i, v_j\}$ is an edge iff $w_{ij} > 0$. The corresponding (undirected) graph (V, E) with $E = \{\{v_i, v_j\} \mid w_{ij} > 0\}$, is called the *underlying graph* of G .

Remark: Since $w_{ii} = 0$, these graphs have no self-loops. We can think of the matrix W as a generalized adjacency matrix. The case where $w_{ij} \in \{0, 1\}$ is equivalent to the notion of a graph as in Definition 1.4.

We can think of the weight w_{ij} of an edge $\{v_i, v_j\}$ as a degree of similarity (or affinity) in an image, or a cost in a network.

An example of a weighted graph is shown in Figure 1.3. The thickness of an edge corresponds to the magnitude of its weight.

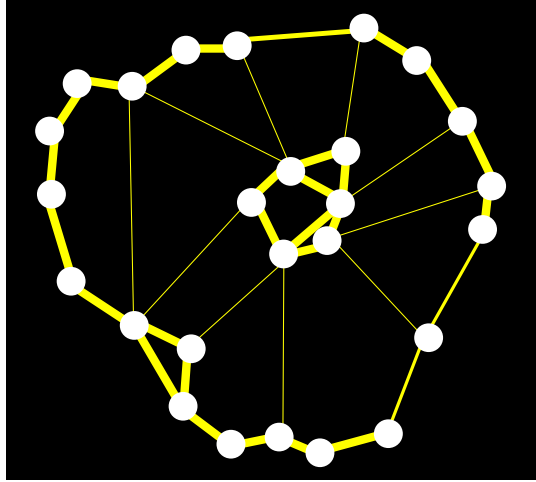


Figure 1.3: A weighted graph.

For every node $v_i \in V$, the *degree* $d(v_i)$ of v_i is the sum of the weights of the edges adjacent to v_i :

$$d(v_i) = \sum_{j=1}^m w_{ij}.$$

Note that in the above sum, only nodes v_j such that there is an edge $\{v_i, v_j\}$ have a nonzero contribution. Such nodes are said to be *adjacent* to v_i , and we write $v_i \sim v_j$.

The degree matrix D is defined as before, namely by $D = \text{diag}(d(v_1), \dots, d(v_m))$.

The weight matrix W can be viewed as a linear map from $\mathbb{R}^V$ to itself. For all $x \in \mathbb{R}^m$, we have

$$(Wx)_i = \sum_{j \sim i} w_{ij}x_j;$$

that is, the value of Wx at v_i is the weighted sum of the values of x at the nodes v_j adjacent to v_i .

Observe that $W\mathbf{1}$ is the (column) vector $(d(v_1), \dots, d(v_m))$ consisting of the degrees of the nodes of the graph.

Given any subset of nodes $A \subseteq V$, we define the *volume* $\text{vol}(A)$ of A as the sum of the weights of all edges adjacent to nodes in A :

$$\text{vol}(A) = \sum_{v_i \in A} d(v_i) = \sum_{v_i \in A} \sum_{j=1}^m w_{ij}.$$

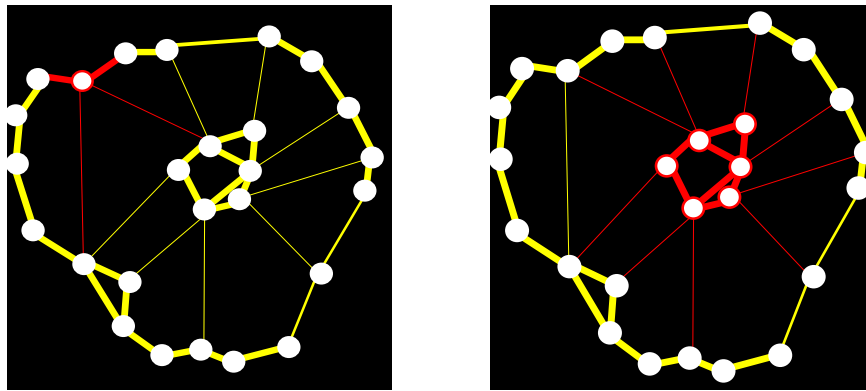


Figure 1.4: Degree and volume.

Observe that $\text{vol}(A) = 0$ if A consists of isolated vertices, that is, if $w_{ij} = 0$ for all $v_i \in A$. Thus, it is best to assume that G does not have isolated vertices.

Given any two subset $A, B \subseteq V$ (not necessarily distinct), we define $\text{links}(A, B)$ by

$$\text{links}(A, B) = \sum_{v_i \in A, v_j \in B} w_{ij}.$$

Since the matrix W is symmetric, we have

$$\text{links}(A, B) = \text{links}(B, A),$$

and observe that $\text{vol}(A) = \text{links}(A, V)$.

The quantity $\text{links}(A, \overline{A}) = \text{links}(\overline{A}, A)$, where $\overline{A} = V - A$ denotes the complement of A in V , measures *how many links escape from A (and $\overline{A}$)*, and the quantity $\text{links}(A, A)$ measures *how many links stay within A itself*.

The quantity

$$\text{cut}(A) = \text{links}(A, \overline{A})$$

is often called the *cut* of A , and the quantity

$$\text{assoc}(A) = \text{links}(A, A)$$

is often called the *association* of A . Clearly,

$$\text{cut}(A) + \text{assoc}(A) = \text{vol}(A).$$

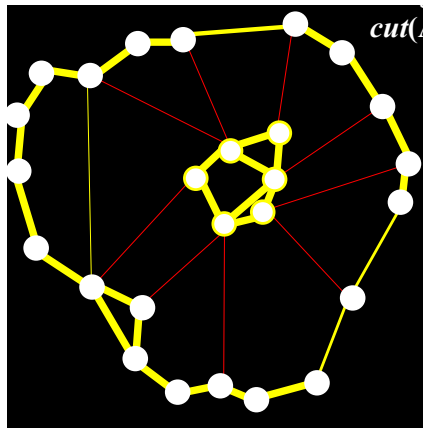


Figure 1.5: A Cut involving the set of nodes in the center and the nodes on the perimeter.

We now define the most important concept of these notes: The Laplacian matrix of a graph. Actually, as we will see, it comes in several flavors.

1.2 Laplacian Matrices of Graphs

Let us begin with directed graphs, although as we will see, graph Laplacians are fundamentally associated with undirected graph.

The key proposition below shows how $BB^\top$ relates to the adjacency matrix A . We reproduce the proof in Gallier [5] (see also Godsil and Royle [7]).

Proposition 1.2. *Given any directed graph G if B is the incidence matrix of G , A is the adjacency matrix of G , and D is the degree matrix such that $D_{ii} = d(v_i)$, then*

$$BB^\top = D - A.$$

Consequently, $BB^\top$ is independent of the orientation of G and $D - A$ is symmetric, positive, semidefnite; that is, the eigenvalues of $D - A$ are real and nonnegative.

The matrix $L = BB^\top = D - A$ is called the *(unnormalized) graph Laplacian* of the graph G .

For example, the graph Laplacian of graph G_1 is

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 4 & -1 & -1 & -1 \\ -1 & -1 & 3 & -1 & 0 \\ 0 & -1 & -1 & 3 & -1 \\ 0 & -1 & 0 & -1 & 2 \end{pmatrix}.$$

The *(unnormalized) graph Laplacian* of an undirected graph $G = (V, E)$ is defined by

$$L = D - A.$$

Observe that each row of L sums to zero (because $B^\top \mathbf{1} = 0$). Consequently, the vector $\mathbf{1}$ is in the nullspace of L .

Remark: With the unoriented version of the incidence matrix (see Definition 1.6), it can be shown that

$$BB^\top = D + A.$$

The natural generalization of the notion of graph Laplacian to weighted graphs is this:

Definition 1.10. Given any weighted graph $G = (V, W)$ with $V = \{v_1, \dots, v_m\}$, the (*unnormalized*) *graph Laplacian* $L(G)$ of G is defined by

$$L(G) = D(G) - W,$$

where $D(G) = \text{diag}(d_1, \dots, d_m)$ is the degree matrix of G (a diagonal matrix), with

$$d_i = \sum_{j=1}^m w_{ij}.$$

As usual, unless confusion arises, we write L instead of $L(G)$.

The graph Laplacian can be interpreted as a linear map from $\mathbb{R}^V$ to itself. For all $x \in \mathbb{R}^V$, we have

$$(Lx)_i = \sum_{j \sim i} w_{ij}(x_i - x_j).$$

It is clear that each row of L sums to 0, so the vector $\mathbf{1}$ is in the nullspace of L , but it is less obvious that L is positive semidefinite. One way to prove it is to generalize slightly the notion of incidence matrix.

Definition 1.11. Given a weighted graph $G = (V, W)$, with $V = \{v_1, \dots, v_m\}$, if $\{e_1, \dots, e_n\}$ are the edges of the underlying graph of G (recall that $\{v_i, v_j\}$ is an edge of this graph iff $w_{ij} > 0$), for any oriented graph G^σ obtained by giving an orientation to the underlying graph of G , the *incidence matrix* B^σ of G^σ is the $m \times n$ matrix whose entries b_{ij} are given by

$$b_{ij} = \begin{cases} +\sqrt{w_{ij}} & \text{if } s(e_j) = v_i \\ -\sqrt{w_{ij}} & \text{if } t(e_j) = v_i \\ 0 & \text{otherwise.} \end{cases}$$

For example, given the weight matrix

$$W = \begin{pmatrix} 0 & 3 & 6 & 3 \\ 3 & 0 & 0 & 3 \\ 6 & 0 & 0 & 3 \\ 3 & 3 & 3 & 0 \end{pmatrix},$$

the incidence matrix B corresponding to the orientation of the underlying graph of W where an edge (i, j) is oriented positively iff $i < j$ is

$$B = \begin{pmatrix} 1.7321 & 2.4495 & 1.7321 & 0 & 0 \\ -1.7321 & 0 & 0 & 1.7321 & 0 \\ 0 & -2.4495 & 0 & 0 & 1.7321 \\ 0 & 0 & -1.7321 & -1.7321 & -1.7321 \end{pmatrix}.$$

The reader should verify that $BB^\top = D - W$. This is true in general, see Proposition 1.3.

It is easy to see that Proposition 1.1 applies to the underlying graph of G .

For any oriented graph G^σ obtained from the underlying graph of G , the rank of the incidence matrix B^σ is equal to $m - c$, where c is the number of connected components of the underlying graph of G , and we have $(B^\sigma)^\top \mathbf{1} = 0$.

We also have the following version of Proposition 1.2 whose proof is immediately adapted.

Proposition 1.3. *Given any weighted graph $G = (V, W)$ with $V = \{v_1, \dots, v_m\}$, if B^σ is the incidence matrix of any oriented graph G^σ obtained from the underlying graph of G and D is the degree matrix of W , then*

$$B^\sigma(B^\sigma)^\top = D - W = L.$$

Consequently, $B^\sigma(B^\sigma)^\top$ is independent of the orientation of the underlying graph of G and $L = D - W$ is symmetric, positive, semidefinite; that is, the eigenvalues of $L = D - W$ are real and nonnegative.

Another way to prove that L is positive semidefinite is to evaluate the quadratic form $x^\top Lx$.

Proposition 1.4. *For any $m \times m$ symmetric matrix $W = (w_{ij})$, if we let $L = D - W$ where D is the degree matrix associated with W , then we have*

$$x^\top Lx = \frac{1}{2} \sum_{i,j=1}^m w_{ij}(x_i - x_j)^2 \quad \text{for all } x \in \mathbb{R}^m.$$

Consequently, $x^\top Lx$ does not depend on the diagonal entries in W , and if $w_{ij} \geq 0$ for all $i, j \in \{1, \dots, m\}$, then L is positive semidefinite.

Proposition 1.4 immediately implies the following facts:
For any weighted graph $G = (V, W)$,

1. The eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_m$ of L are real and nonnegative, and there is an orthonormal basis of eigenvectors of L .
2. The smallest eigenvalue λ_1 of L is equal to 0, and $\mathbf{1}$ is a corresponding eigenvector.

It turns out that the dimension of the nullspace of L (the eigenspace of 0) is equal to the number of connected components of the underlying graph of G . This is an immediate consequence of Proposition 1.1 and the fact that $L = BB^\top$.

Proposition 1.5. *Let $G = (V, W)$ be a weighted graph. The number c of connected components $K_1, \dots, K_c$ of the underlying graph of G is equal to the dimension of the nullspace of L , which is equal to the multiplicity of the eigenvalue 0. Furthermore, the nullspace of L has a basis consisting of indicator vectors of the connected components of G , that is, vectors $(f_1, \dots, f_m)$ such that $f_j = 1$ iff $v_j \in K_i$ and $f_j = 0$ otherwise.*

Proposition 1.5 implies that if the underlying graph of G is connected, then the second eigenvalue λ_2 of L is strictly positive.

Remarkably, the eigenvalue λ_2 contains a lot of information about the graph G (assuming that $G = (V, E)$ is an undirected graph).

This was first discovered by Fiedler in 1973, and for this reason, λ_2 is often referred to as the *Fiedler number*.

For more on the properties of the Fiedler number, see Godsil and Royle [7] (Chapter 13) and Chung [3].

More generally, the spectrum $(0, \lambda_2, \dots, \lambda_m)$ of L contains a lot of information about the combinatorial structure of the graph G . Leverage of this information is the object of *spectral graph theory*.

It turns out that normalized variants of the graph Laplacian are needed, especially in applications to graph clustering.

These variants make sense only if G has no isolated vertices, which means that every row of W contains some strictly positive entry.

In this case, the degree matrix D contains positive entries, so it is invertible and $D^{-1/2}$ makes sense; namely

$$D^{-1/2} = \text{diag}(d_1^{-1/2}, \dots, d_m^{-1/2}),$$

and similarly for any real exponent α .

Definition 1.12. Given any weighted directed graph $G = (V, W)$ with no isolated vertex and with $V = \{v_1, \dots, v_m\}$, the *(normalized) graph Laplacians L_{sym} and L_{rw} of G* are defined by

$$\begin{aligned} L_{\text{sym}} &= D^{-1/2} L D^{-1/2} = I - D^{-1/2} W D^{-1/2} \\ L_{\text{rw}} &= D^{-1} L = I - D^{-1} W. \end{aligned}$$

Observe that the Laplacian $L_{\text{sym}} = D^{-1/2} L D^{-1/2}$ is a symmetric matrix (because L and $D^{-1/2}$ are symmetric) and that

$$L_{\text{rw}} = D^{-1/2} L_{\text{sym}} D^{1/2}.$$

The reason for the notation L_{rw} is that this matrix is closely related to a random walk on the graph G .

Since the unnormalized Laplacian L can be written as $L = BB^\top$, where B is the incidence matrix of any oriented graph obtained from the underlying graph of $G = (V, W)$, if we let

$$B_{\text{sym}} = D^{-1/2}B,$$

we get

$$L_{\text{sym}} = B_{\text{sym}}B_{\text{sym}}^\top.$$

In particular, for any singular decomposition $B_{\text{sym}} = U\Sigma V^\top$ of B_{sym} (with U an $m \times m$ orthogonal matrix, Σ a “diagonal” $m \times n$ matrix of singular values, and V an $n \times n$ orthogonal matrix), the eigenvalues of L_{sym} are the squares of the top m singular values of B_{sym} , and the vectors in U are orthonormal eigenvectors of L_{sym} with respect to these eigenvalues (the squares of the top m diagonal entries of Σ).

Computing the SVD of B_{sym} generally yields more accurate results than diagonalizing L_{sym} , especially when L_{sym} has eigenvalues with high multiplicity.

Proposition 1.6. *Let $G = (V, W)$ be a weighted graph without isolated vertices. The graph Laplacians, L , L_{sym} , and L_{rw} satisfy the following properties:*

- (1) *The matrix L_{sym} is symmetric, positive, semidefinite. In fact,*

$$x^\top L_{\text{sym}} x = \frac{1}{2} \sum_{i,j=1}^m w_{ij} \left(\frac{x_i}{\sqrt{d_i}} - \frac{x_j}{\sqrt{d_j}} \right)^2 \quad x \in \mathbb{R}^m.$$

- (2) *The normalized graph Laplacians L_{sym} and L_{rw} have the same spectrum*

($0 = \nu_1 \leq \nu_2 \leq \dots \leq \nu_m$), and a vector $u \neq 0$ is an eigenvector of L_{rw} for λ iff $D^{1/2}u$ is an eigenvector of L_{sym} for λ .

- (3) *The graph Laplacians, L , L_{sym} , and L_{rw} are symmetric, positive, semidefinite.*

- (4) *A vector $u \neq 0$ is a solution of the generalized eigenvalue problem $Lu = \lambda Du$ iff $D^{1/2}u$ is an eigenvector of L_{sym} for the eigenvalue λ iff u is an eigenvector of L_{rw} for the eigenvalue λ .*

- (5) *The graph Laplacians, L and L_{rw} have the same nullspace. For any vector u , we have $u \in \text{Ker}(L)$ iff $D^{1/2}u \in \text{Ker}(L_{\text{sym}})$.*
- (6) *The vector $\mathbf{1}$ is in the nullspace of L_{rw} , and $D^{1/2}\mathbf{1}$ is in the nullspace of L_{sym} .*
- (7) *For every eigenvalue ν_i of the normalized graph Laplacian L_{sym} , we have $0 \leq \nu_i \leq 2$. Furthermore, $\nu_m = 2$ iff the underlying graph of G contains a nontrivial connected bipartite component.*
- (8) *If $m \geq 2$ and if the underlying graph of G is not a complete graph, then $\nu_2 \leq 1$. Furthermore the underlying graph of G is a complete graph iff $\nu_2 = \frac{m}{m-1}$.*
- (9) *If $m \geq 2$ and if the underlying graph of G is connected then $\nu_2 > 0$.*
- (10) *If $m \geq 2$ and if the underlying graph of G has no isolated vertices, then $\nu_m \geq \frac{m}{m-1}$.*

A version of Proposition 1.5 also holds for the graph Laplacians L_{sym} and L_{rw} .

This follows easily from the fact that Proposition 1.1 applies to the underlying graph of a weighted graph. The proof is left as an exercise.

Proposition 1.7. *Let $G = (V, W)$ be a weighted graph. The number c of connected components $K_1, \dots, K_c$ of the underlying graph of G is equal to the dimension of the nullspace of both L_{sym} and L_{rw} , which is equal to the multiplicity of the eigenvalue 0. Furthermore, the nullspace of L_{rw} has a basis consisting of indicator vectors of the connected components of G , that is, vectors $(f_1, \dots, f_m)$ such that $f_j = 1$ iff $v_j \in K_i$ and $f_j = 0$ otherwise. For L_{sym} , a basis of the nullspace is obtained by multiplying the above basis of the nullspace of L_{rw} by $D^{1/2}$.*

Chapter 2

Spectral Graph Drawing

2.1 Graph Drawing and Energy Minimization

Let $G = (V, E)$ be some undirected graph. It is often desirable to draw a graph, usually in the plane but possibly in 3D, and it turns out that the graph Laplacian can be used to design surprisingly good methods.

Say $|V| = m$. The idea is to assign a point $\rho(v_i)$ in $\mathbb{R}^n$ to the vertex $v_i \in V$, for every $v_i \in V$, and to draw a line segment between the points $\rho(v_i)$ and $\rho(v_j)$.

Thus, a *graph drawing* is a function $\rho: V \rightarrow \mathbb{R}^n$.

We define the *matrix of a graph drawing ρ (in $\mathbb{R}^n$)* as a $m \times n$ matrix R whose i th row consists of the row vector $\rho(v_i)$ corresponding to the point representing v_i in $\mathbb{R}^n$.

Typically, we want $n < m$; in fact n should be much smaller than m .

A representation is *balanced* iff the sum of the entries of every column is zero, that is,

$$\mathbf{1}^\top R = 0.$$

If a representation is not balanced, it can be made balanced by a suitable translation.

We may also assume that the columns of R are linearly independent, since any basis of the column space also determines the drawing. Thus, from now on, we may assume that $n \leq m$.

Remark: A graph drawing $\rho: V \rightarrow \mathbb{R}^n$ is not required to be injective, which may result in degenerate drawings where distinct vertices are drawn as the same point.

For this reason, we prefer not to use the terminology *graph embedding*, which is often used in the literature. This is because in differential geometry, an embedding always refers to an injective map.

The term *graph immersion* would be more appropriate.

As explained in Godsil and Royle [7], we can imagine building a physical model of G by connecting adjacent vertices (in $\mathbb{R}^n$) by identical springs.

Then, it is natural to consider a *representation to be better if it requires the springs to be less extended*.

We can formalize this by defining the *energy* of a drawing R by

$$\mathcal{E}(R) = \sum_{\{v_i, v_j\} \in E} \|\rho(v_i) - \rho(v_j)\|^2,$$

where $\rho(v_i)$ is the i th row of R and $\|\rho(v_i) - \rho(v_j)\|^2$ is the square of the Euclidean length of the line segment joining $\rho(v_i)$ and $\rho(v_j)$.

Then, “good drawings” are drawings that minimize the energy function $\mathcal{E}$.

Of course, the trivial representation corresponding to the zero matrix is optimum, so we need to impose extra constraints to rule out the trivial solution.

We can consider the more general situation where the springs are not necessarily identical. This can be modeled by a symmetric weight (or stiffness) matrix $W = (w_{ij})$, with $w_{ij} \geq 0$.

Then our energy function becomes

$$\mathcal{E}(R) = \sum_{\{v_i, v_j\} \in E} w_{ij} \|\rho(v_i) - \rho(v_j)\|^2.$$

It turns out that this function can be expressed in terms of the Laplacian $L = D - W$.

Proposition 2.1. *Let $G = (V, W)$ be a weighted graph, with $|V| = m$ and W an $m \times m$ symmetric matrix, and let R be the matrix of a graph drawing ρ of G in $\mathbb{R}^n$ (a $m \times n$ matrix). If $L = D - W$ is the unnormalized Laplacian matrix associated with W , then*

$$\mathcal{E}(R) = \text{tr}(R^\top L R).$$

Since the matrix $R^\top LR$ is symmetric, it has real eigenvalues. Actually, since L is positive semidefinite, so is $R^\top LR$.

Then, the trace of $R^\top LR$ is equal to the sum of its positive eigenvalues, and this is the energy $\mathcal{E}(R)$ of the graph drawing.

If R is the matrix of a graph drawing in $\mathbb{R}^n$, then for any invertible matrix M , the map that assigns $\rho(v_i)M$ to v_i is another graph drawing of G , and these two drawings convey the same amount of information.

From this point of view, a graph drawing is determined by the column space of R . Therefore, it is reasonable to assume that the columns of R are pairwise orthogonal and that they have unit length.

Such a matrix satisfies the equation $R^\top R = I$, and the corresponding drawing is called an *orthogonal drawing*. This condition also rules out trivial drawings.

The following result tells us how to find minimum energy graph drawings, provided the graph is connected.

Theorem 2.2. *Let $G = (V, W)$ be a weighted graph with $|V| = m$. If $L = D - W$ is the (unnormalized) Laplacian of G , and if the eigenvalues of L are $0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_m$, then the minimal energy of any balanced orthogonal graph drawing of G in $\mathbb{R}^n$ is equal to $\lambda_2 + \dots + \lambda_{n+1}$ (in particular, this implies that $n < m$). The $m \times n$ matrix R consisting of any unit eigenvectors $u_2, \dots, u_{n+1}$ associated with $\lambda_2 \leq \dots \leq \lambda_{n+1}$ yields a balanced orthogonal graph drawing of minimal energy; it satisfies the condition $R^\top R = I$.*

Observe that for any orthogonal $n \times n$ matrix Q , since

$$\text{tr}(R^\top LR) = \text{tr}(Q^\top R^\top LRQ),$$

the matrix RQ also yields a minimum orthogonal graph drawing.

Since $\mathbf{1}$ spans the nullspace of L , using u_1 (which belongs to $\text{Ker } L$) as one of the vectors in R would have the effect that all points representing vertices of G would have the same first coordinate.

This would mean that the drawing lives in a hyperplane in $\mathbb{R}^n$, which is undesirable, especially when $n = 2$, where all vertices would be collinear. This is why we omit the first eigenvector u_1 .

In summary, if $\lambda_2 > 0$, an automatic method for drawing a graph in $\mathbb{R}^2$ is this:

1. Compute the two smallest nonzero eigenvalues $\lambda_2 \leq \lambda_3$ of the graph Laplacian L (it is possible that $\lambda_3 = \lambda_2$ if λ_2 is a multiple eigenvalue);
2. Compute two unit eigenvectors u_2, u_3 associated with λ_2 and λ_3 , and let $R = [u_2 \ u_3]$ be the $m \times 2$ matrix having u_2 and u_3 as columns.
3. Place vertex v_i at the point whose coordinates is the i th row of R , that is, (R_{i1}, R_{i2}) .

This method generally gives pleasing results, but beware that there is no guarantee that distinct nodes are assigned distinct images, because R can have identical rows.

2.2 Examples of Graph Drawings

We now give a number of examples using **Matlab**. Some of these are borrowed or adapted from Spielman [14].

Example 1. Consider the graph with four nodes whose adjacency matrix is

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

We use the following program to compute u_2 and u_3 :

```
A = [0 1 1 0; 1 0 0 1; 1 0 0 1; 0 1 1 0];
D = diag(sum(A));
L = D - A;
[v, e] = eigs(L);
gplot(A, v(:, [3 2]))
hold on;
gplot(A, v(:, [3 2]), 'o')
```

The graph of Example 1 is shown in Figure 2.1. It turns out that $\lambda_2 = \lambda_3 = 2$ is a double eigenvalue.

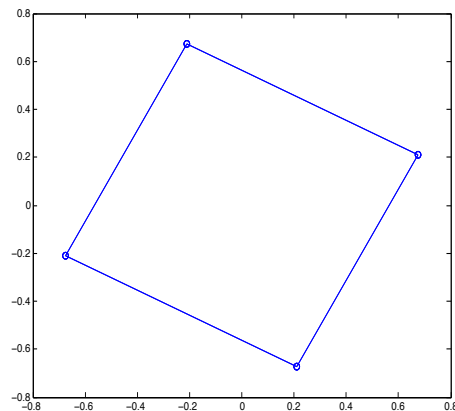


Figure 2.1: Drawing of the graph from Example 1.

Example 2. Consider the graph G_2 shown in Figure 1.2 given by the adjacency matrix

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}.$$

We use the following program to compute u_2 and u_3 :

```

A = [0 1 1 0 0; 1 0 1 1 1; 1 1 0 1 0;
      0 1 1 0 1; 0 1 0 1 0];
D = diag(sum(A));
L = D - A;
[v, e] = eig(L);
gplot(A, v(:, [2 3]))
hold on
gplot(A, v(:, [2 3]), 'o')

```

Note that node v_2 is assigned to the point $(0, 0)$, so the difference between this drawing and the drawing in Figure 1.2 is that the drawing of Figure 2.2 is not convex.

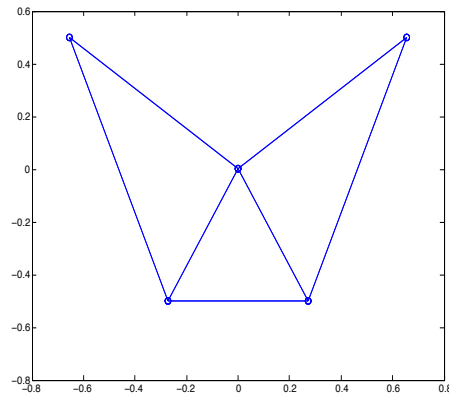


Figure 2.2: Drawing of the graph from Example 2.

Example 3. Consider the ring graph defined by the adjacency matrix A given in the **Matlab** program shown below:

```
A = diag(ones(1, 11),1);
A = A + A';
A(1, 12) = 1; A(12, 1) = 1;
D = diag(sum(A));
L = D - A;
[v, e] = eig(L);
gplot(A, v(:, [2 3]))
hold on
gplot(A, v(:, [2 3]), 'o')
```

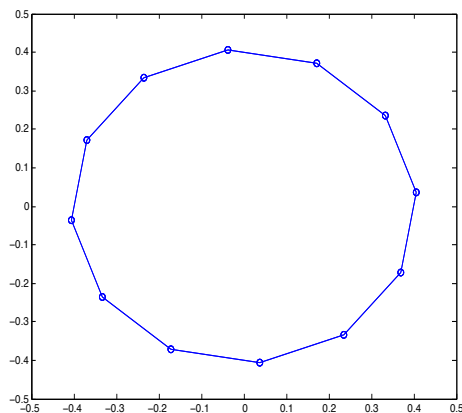


Figure 2.3: Drawing of the graph from Example 3.

Again $\lambda_2 = 0.2679$ is a double eigenvalue (and so are the next pairs of eigenvalues, except the last, $\lambda_{12} = 4$).

Example 4. In this example adapted from Spielman, we generate 20 randomly chosen points in the unit square, compute their Delaunay triangulation, then the adjacency matrix of the corresponding graph, and finally draw the graph using the second and third eigenvalues of the Laplacian.

```
A = zeros(20,20);
xy = rand(20, 2);
trigs = delaunay(xy(:,1), xy(:,2));
elemtrig = ones(3) - eye(3);
for i = 1:length(trigs),
    A(trigs(i,:),trigs(i,:)) = elemtrig;
end
A = double(A > 0);
gplot(A,xy)
D = diag(sum(A));
L = D - A;
[v, e] = eigs(L, 3, 'sm');
figure(2)
gplot(A, v(:, [2 1]))
hold on
gplot(A, v(:, [2 1]), 'o')
```

The Delaunay triangulation of the set of 20 points and the drawing of the corresponding graph are shown in Figure 2.4.

The graph drawing on the right looks nicer than the graph on the left but is no longer planar.

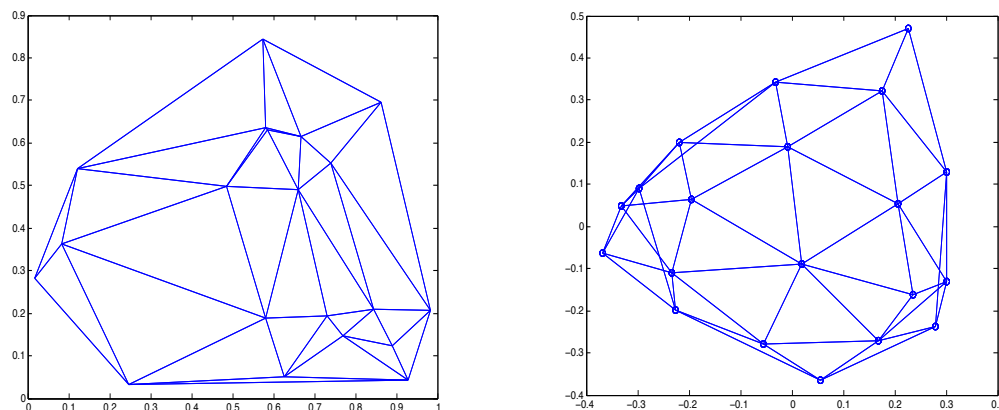


Figure 2.4: Delaunay triangulation (left) and drawing of the graph from Example 4 (right).

Example 5. Our last example, also borrowed from Spielman [14], corresponds to the skeleton of the “Buckyball,” a geodesic dome invented by the architect Richard Buckminster Fuller (1895–1983).

The Montréal Biosphère is an example of a geodesic dome designed by Buckminster Fuller.

```
A = full(bucky);  
D = diag(sum(A));  
L = D - A;  
[v, e] = eig(L);  
gplot(A, v(:, [2 3]))  
hold on;  
gplot(A, v(:, [2 3]), 'o')
```

Figure 2.5 shows a graph drawing of the Buckyball. This picture seems a bit squashed for two reasons. First, it is really a 3-dimensional graph; second, $\lambda_2 = 0.2434$ is a triple eigenvalue. (Actually, the Laplacian of L has many multiple eigenvalues.) What we should really do is to plot this graph in $\mathbb{R}^3$ using three orthonormal eigenvectors associated with λ_2 .

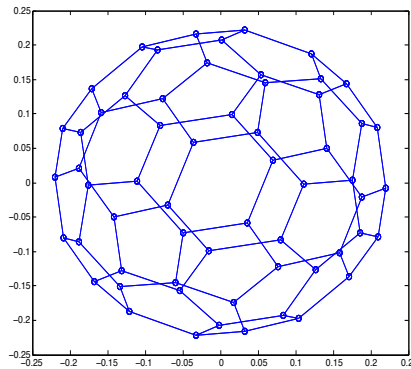


Figure 2.5: Drawing of the graph of the Buckyball.

A 3D picture of the graph of the Buckyball is produced by the following **Matlab** program, and its image is shown in Figure 2.6. It looks better!

```
[x, y] = gplot(A, v(:, [2 3]));
[x, z] = gplot(A, v(:, [2 4]));
plot3(x,y,z)
```

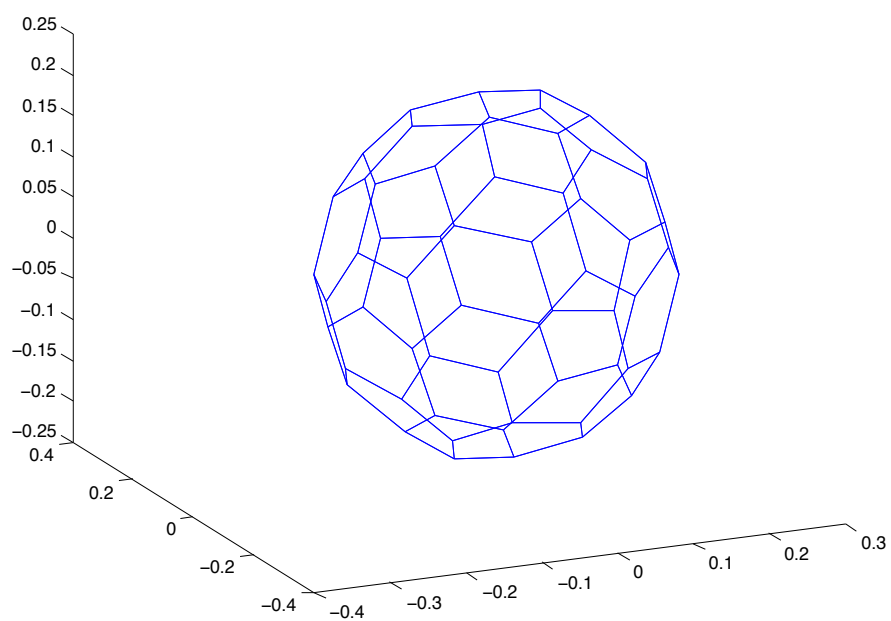


Figure 2.6: Drawing of the graph of the Buckyball in $\mathbb{R}^3$.

Chapter 3

Graph Clustering

3.1 Graph Clustering Using Normalized Cuts

Given a set of data, the goal of clustering is to partition the data into different groups according to their similarities.

When the data is given in terms of a similarity graph G , where the weight w_{ij} between two nodes v_i and v_j is a measure of similarity of v_i and v_j , the problem can be stated as follows:

Find a partition $(A_1, \dots, A_K)$ of the set of nodes V into different groups such that the *edges between different groups have very low weight* (which indicates that the *points in different clusters are dissimilar*), and the *edges within a group have high weight* (which indicates that *points within the same cluster are similar*).

The above graph clustering problem can be formalized as an optimization problem, using the notion of *cut* mentioned at the end of Section 1.1.

Given a subset A of the set of vertices V , we define $\text{cut}(A)$ by

$$\text{cut}(A) = \text{links}(A, \overline{A}) = \sum_{v_i \in A, v_j \in \overline{A}} w_{ij}.$$

If we want to partition V into K clusters, we can do so by finding a partition $(A_1, \dots, A_K)$ that minimizes the quantity

$$\text{cut}(A_1, \dots, A_K) = \frac{1}{2} \sum_{i=1}^K \text{cut}(A_i).$$

For $K = 2$, the mincut problem is a classical problem that can be solved efficiently, but in practice, it does not yield satisfactory partitions.

Indeed, in many cases, the mincut solution separates one vertex from the rest of the graph.

What we need is to design our cost function in such a way that it keeps the subsets A_i “reasonably large” (*reasonably balanced*).

An example of a weighted graph and a partition of its nodes into two clusters is shown in Figure 3.1.

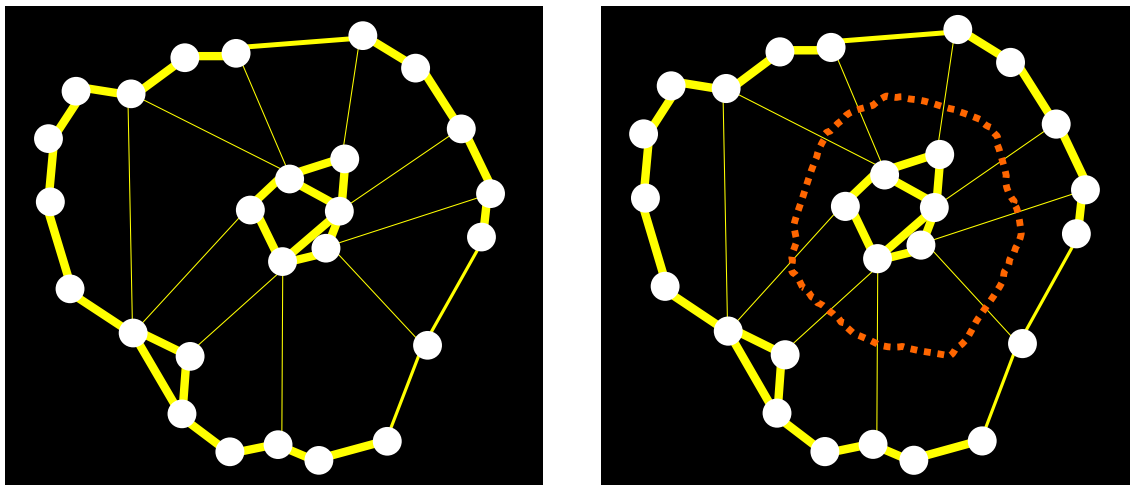


Figure 3.1: A weighted graph and its partition into two clusters.

A way to get around this problem is to *normalize the cuts* by dividing by some measure of each subset A_i .

One possibility is to use the *size* (the number of elements) of A_i .

Another is to use the *volume* $\text{vol}(A_i)$ of A_i .

A solution using the second measure (the volume) (for $K = 2$) was proposed and investigated in a seminal paper of Shi and Malik [13].

Subsequently, Yu (in her dissertation [16]) and Yu and Shi [17] extended the method to $K > 2$ clusters.

The idea is to minimize the cost function

$$\text{Ncut}(A_1, \dots, A_K) = \sum_{i=1}^K \frac{\text{links}(A_i, \overline{A_i})}{\text{vol}(A_i)} = \sum_{i=1}^K \frac{\text{cut}(A_i, \overline{A_i})}{\text{vol}(A_i)}.$$

We begin with the case $K = 2$, which is easier to handle.

3.2 Special Case: 2-Way Clustering Using Normalized Cuts

Our goal is to express our optimization problem in matrix form.

In the case of two clusters, a single vector X can be used to describe the partition $(A_1, A_2) = (A, \overline{A})$.

We need to choose the structure of this vector in such a way that $\text{Ncut}(A, \overline{A})$ is equal to the Rayleigh ratio

$$\frac{X^\top L X}{X^\top D X}.$$

It is also important to pick a *vector representation which is invariant under multiplication by a nonzero scalar*, because the Rayleigh ratio is scale-invariant, and it is crucial to take advantage of this fact to make the denominator go away.

Let $N = |V|$ be the number of nodes in the graph G .

In view of the desire for a scale-invariant representation, it is natural to assume that the vector X is of the form

$$X = (x_1, \dots, x_N),$$

where $x_i \in \{a, b\}$ for $i = 1, \dots, N$, for any two distinct real numbers a, b .

This is an indicator vector in the sense that, for $i = 1, \dots, N$,

$$x_i = \begin{cases} a & \text{if } v_i \in A \\ b & \text{if } v_i \notin A. \end{cases}$$

The correct interpretation is really to view X as a representative of a point in the *real projective space* $\mathbb{RP}^{N-1}$, namely the point $\mathbb{P}(X)$ of homogeneous coordinates $(x_1 : \cdots : x_N)$.

Therefore, from now on, we view X as a vector of homogeneous coordinates representing the point $\mathbb{P}(X) \in \mathbb{RP}^{N-1}$.

Let $d = \mathbf{1}^\top D \mathbf{1}$ and $\alpha = \text{vol}(A)$. Then, $\text{vol}(\overline{A}) = d - \alpha$.

Using Proposition 1.4, we have

$$\begin{aligned} X^\top L X &= (a - b)^2 \text{cut}(A, \overline{A}) \\ X^\top D X &= \alpha a^2 + (d - \alpha) b^2. \end{aligned}$$

We obtain

$$\text{Ncut}(A, \overline{A}) = \frac{d}{\alpha(d - \alpha)} \text{cut}(A, \overline{A})$$

and

$$\frac{X^\top LX}{X^\top DX} = \frac{(a - b)^2}{\alpha a^2 + (d - \alpha)b^2} \text{cut}(A, \overline{A}).$$

In order to have

$$\text{Ncut}(A, \overline{A}) = \frac{X^\top LX}{X^\top DX},$$

we need to find a and b so that

$$\frac{(a - b)^2}{\alpha a^2 + (d - \alpha)b^2} = \frac{d}{\alpha(d - \alpha)}.$$

We find the condition

$$a\alpha + b(d - \alpha) = 0. \quad (\dagger)$$

Note that condition $(\dagger)$ applied to a vector X whose components are a or b is equivalent to the fact that X is orthogonal to $D\mathbf{1}$.

Various choices for a choice of scale factor appear in the literature.

von Luxburg [15] picks

$$a = \sqrt{\frac{d - \alpha}{\alpha}}, \quad b = -\sqrt{\frac{\alpha}{d - \alpha}}.$$

Shi and Malik [13] use

$$a = 1, \quad b = -\frac{\alpha}{d - \alpha} = -\frac{k}{1 - k},$$

with

$$k = \frac{\alpha}{d}.$$

Belkin and Niyogi [2] use

$$a = \frac{1}{\alpha}, \quad b = -\frac{1}{d - \alpha}.$$

However, there is no need to restrict solutions to be of either of these forms.

So, let

$$\mathcal{X} = \{(x_1, \dots, x_N) \mid x_i \in \{a, b\}, a, b \in \mathbb{R}, a, b \neq 0\},$$

so that our solution set is

$$\mathcal{K} = \{X \in \mathcal{X} \mid X^\top D \mathbf{1} = 0\},$$

Actually, to be perfectly rigorous, we are looking for solutions in $\mathbb{RP}^{N-1}$, so our solution set is really

$$\mathbb{P}(\mathcal{K}) = \{(x_1 : \dots : x_N) \in \mathbb{RP}^{N-1} \mid (x_1, \dots, x_N) \in \mathcal{K}\}.$$

Consequently, our minimization problem can be stated as follows:

Problem PNC1

$$\begin{array}{ll}
\text{minimize} & \frac{X^\top L X}{X^\top D X} \\
\text{subject to} & X^\top D \mathbf{1} = 0, \quad X \in \mathcal{X}.
\end{array}$$

It is understood that the solutions are points $\mathbb{P}(X)$ in $\mathbb{RP}^{N-1}$.

Since the Rayleigh ratio and the constraints $X^\top D \mathbf{1} = 0$ and $X \in \mathcal{X}$ are scale-invariant we are led to the following formulation of our problem:

Problem PNC2

$$\begin{aligned} & \text{minimize} && X^\top LX \\ & \text{subject to} && X^\top DX = 1, \quad X^\top D\mathbf{1} = 0, \quad X \in \mathcal{X}. \end{aligned}$$

Problem PNC2 is equivalent to problem PNC1, in the sense that if X is any minimal solution of PNC1, then $X/(X^\top DX)^{1/2}$ is a minimal solution of PNC2 (with the same minimal value for the objective functions), and if X is a minimal solution of PNC2, then λX is a minimal solution for PNC1 for all $\lambda \neq 0$ (with the same minimal value for the objective functions).

Equivalently, problems PNC1 and PNC2 have the same set of minimal solutions as points $\mathbb{P}(X) \in \mathbb{RP}^{N-1}$ given by their homogeneous coordinates X .

Unfortunately, this is an NP-complete problem, as shown by Shi and Malik [13].

As often with hard combinatorial problems, we can look for a *relaxation* of our problem, which means looking for an optimum in a larger continuous domain.

After doing this, the problem is to find a discrete solution which is close to a continuous optimum of the relaxed problem.

The natural relaxation of this problem is to allow X to be any nonzero vector in $\mathbb{R}^N$, and we get the problem:

$$\begin{aligned} \text{minimize} \quad & X^\top L X \\ \text{subject to} \quad & X^\top D X = 1, \quad X^\top D \mathbf{1} = 0. \end{aligned}$$

In order to apply Proposition A.2, we make the change of variable $Y = D^{1/2}X$, so that $X = D^{-1/2}Y$.

We obtain the problem:

$$\begin{aligned} \text{minimize} \quad & Y^\top D^{-1/2} L D^{-1/2} Y \\ \text{subject to} \quad & Y^\top Y = 1, \quad Y^\top D^{1/2} \mathbf{1} = 0. \end{aligned}$$

Because $L\mathbf{1} = 0$, the vector $D^{1/2}\mathbf{1}$ belongs to the nullspace of the symmetric Laplacian $L_{\text{sym}} = D^{-1/2} L D^{-1/2}$.

By Proposition A.2, minima are achieved by any unit eigenvector Y of the second eigenvalue $\nu_2 > 0$ of L_{sym} .

Then, $Z = D^{-1/2}Y$ is a solution of our original relaxed problem

Note that because Z is nonzero and orthogonal to $D\mathbf{1}$, a vector with positive entries, it must have negative and positive entries.

The next question is to figure how close is Z to an exact solution in $\mathcal{X}$.

Actually, because solutions are points in $\mathbb{RP}^{N-1}$, the correct statement of the question is: Find an exact solution $\mathbb{P}(X) \in \mathbb{P}(\mathcal{X})$ which is the closest (in a suitable sense) to the approximate solution $\mathbb{P}(Z) \in \mathbb{RP}^{N-1}$.

However, because $\mathcal{X}$ is closed under the antipodal map, it can be shown that minimizing the distance $d(\mathbb{P}(X), \mathbb{P}(Z))$ on $\mathbb{RP}^{N-1}$ is equivalent to minimizing the Euclidean distance $\|X - Z\|_2$, where X and Z are representatives of $\mathbb{P}(X)$ and $\mathbb{P}(Z)$ on the unit sphere (if we use the Riemannian metric on $\mathbb{RP}^{N-1}$ induced by the Euclidean metric on $\mathbb{R}^N$).

We may assume $b < 0$, in which case $a > 0$.

If all entries in Z are nonzero, due to the projective nature of the solution set, it seems reasonable to say that the partition of V is defined by the signs of the entries in Z .

Thus, A will consist of nodes those v_i for which $x_i > 0$.

Elements corresponding to zero entries can be assigned to either A or $\overline{A}$, unless additional information is available. In our implementation, they are assigned to A .

Here are some examples of normalized cuts found by a fairly naive implementation of the method.

The weight matrix of the first example is

$$W_1 = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{pmatrix}.$$

Its underlying graph has 9 nodes and 9 edges and is shown in Figure 3.2 on the left.

The normalized cut found by the algorithm is shown in the middle; the edge of the cut is shown in magenta, and the vertices of the blocks of the partition are shown in blue and red.

The figure on the right shows the two disjoint subgraphs obtained after deleting the cut edge.

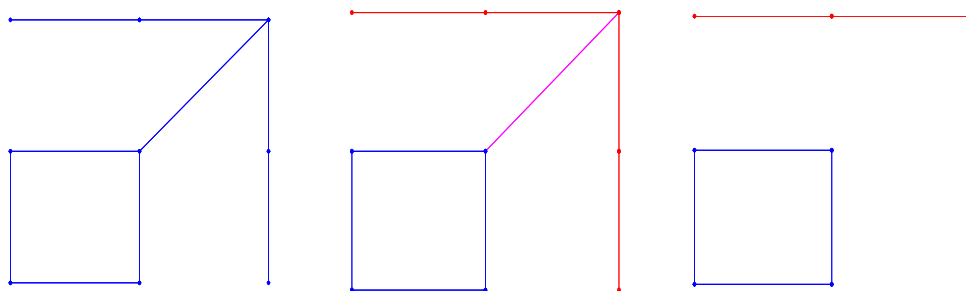


Figure 3.2: Underlying graph of the matrix W_1 (left); normalized cut (middle); blocks of the cut (right).

The weight matrix of the second example is

$$W_2 = \begin{pmatrix} 0 & 3 & 6 & 3 \\ 3 & 0 & 0 & 3 \\ 6 & 0 & 0 & 3 \\ 3 & 3 & 3 & 0 \end{pmatrix}.$$

Its underlying graph has 4 nodes and 5 edges and is shown in Figure 3.3 on the left.

The normalized cut found by the algorithm is shown in the middle; the edges of the cut are shown in magenta, and the vertices of the blocks of the partition are shown in blue and red.

The figure on the right shows the two disjoint subgraphs obtained after deleting the cut edges.

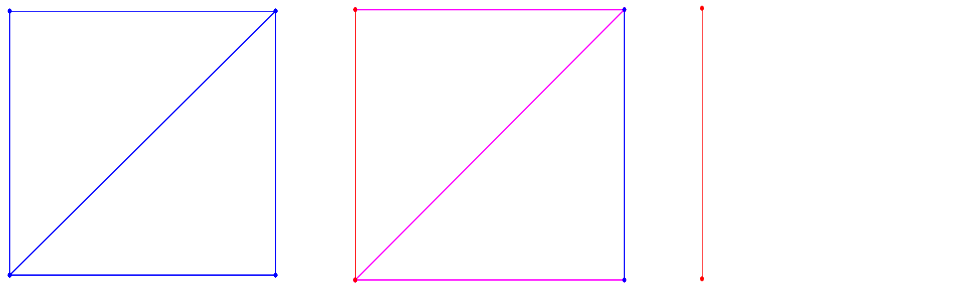


Figure 3.3: Underlying graph of the matrix W_2 (left); normalized cut (middle); blocks of the cut (right).

The weight matrix W_3 of the third example is the adjacency matrix of the complete graph on 12 vertices.

All nondiagonal entries are equal to 1, and the diagonal entries are equal to 0.

This graph has 66 edges and is shown in Figure 3.4 on the left.

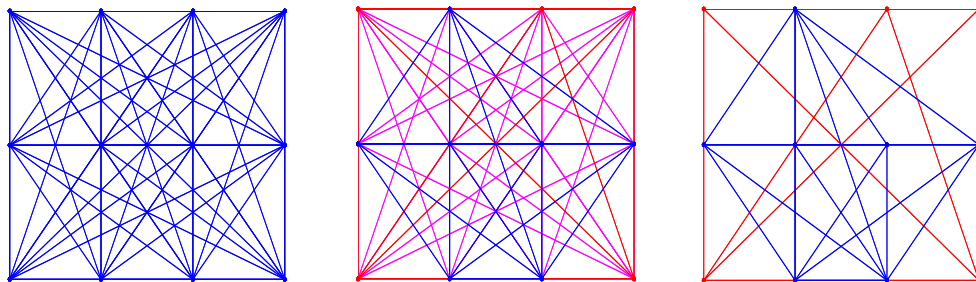


Figure 3.4: Underlying graph of the matrix W_3 (left); normalized cut (middle); blocks of the cut (right).

The normalized cut found by the algorithm is shown in the middle; the edges of the cut are shown in magenta, and the vertices of the blocks of the partition are shown in blue and red.

The figure on the right shows the two disjoint subgraphs obtained after deleting the cut edges.

Our naive algorithm treated zero as a positive entry. Now, using the fact that

$$b = -\frac{\alpha a}{d - \alpha},$$

a better solution is to look for a vector $X \in \mathbb{R}^N$ with $X_i \in \{a, b\}$ which is closest to a minimum Z of the relaxed problem (in the sense that $\|X - Z\|$ is minimized) and with $\|X\| = \|Z\|$.

I implemented such an algorithm, and it seems to do a good job dealing with zero entries in the continuous solution Z .

3.3 *K*-Way Clustering Using Normalized Cuts

We now consider the general case in which $K \geq 3$.

Two crucial issues need to be addressed (to the best of our knowledge, these points are not clearly articulated in the literature).

1. The choice of a matrix representation for partitions on the set of vertices.

It is important that such a representation be scale-invariant.

It is also necessary to state necessary and sufficient conditions for such matrices to represent a partition.

2. The choice of a metric to compare solutions.

We describe a partition $(A_1, \dots, A_K)$ of the set of nodes V by an $N \times K$ matrix $X = [X^1 \dots X^K]$ whose columns $X^1, \dots, X^K$ are indicator vectors of the partition $(A_1, \dots, A_K)$.

Inspired by what we did in Section 3.2, we assume that the vector X^j is of the form

$$X^j = (x_1^j, \dots, x_N^j),$$

where $x_i^j \in \{a_j, b_j\}$ for $j = 1, \dots, K$ and $i = 1, \dots, N$, and where a_j, b_j are any two distinct real numbers.

The vector X^j is an indicator vector for A_j in the sense that, for $i = 1, \dots, N$,

$$x_i^j = \begin{cases} a_j & \text{if } v_i \in A_j \\ b_j & \text{if } v_i \notin A_j. \end{cases}$$

When $\{a_j, b_j\} = \{0, 1\}$ for $j = 1, \dots, K$, such a matrix is called a *partition matrix* by Yu and Shi.

However, such a choice is premature, since it is better to have a scale-invariant representation to make the denominators of the Rayleigh ratios go away.

Since the partition $(A_1, \dots, A_K)$ consists of nonempty pairwise disjoint blocks whose union is V , some conditions on X are required to reflect these properties, but we will worry about this later.

As in Section 3.2, we seek conditions on the a_j s and the b_j s in order to express the normalized cut $\text{Ncut}(A_1, \dots, A_K)$ as a sum of Rayleigh ratios.

Then, we reformulate our optimization problem in a more convenient form, by chasing the denominators in the Rayleigh ratios, and by expressing the objective function in terms of the *trace* of a certain matrix.

This will reveal the important fact that the solutions of the relaxed problem are right-invariant under multiplication by a $K \times K$ orthogonal matrix.

Let $d = \mathbf{1}^\top D \mathbf{1}$ and $\alpha_j = \text{vol}(A_j)$, so that $\alpha_1 + \cdots + \alpha_K = d$.

Then, $\text{vol}(\overline{A_j}) = d - \alpha_j$, and as in Section 3.2, we have

$$\begin{aligned} (X^j)^\top L X^j &= (a_j - b_j)^2 \text{cut}(A_j, \overline{A_j}), \\ (X^j)^\top D X^j &= \alpha_j a_j^2 + (d - \alpha_j) b_j^2. \end{aligned}$$

When $K \geq 3$, unlike the case $K = 2$, in general we have $\text{cut}(A_j, \overline{A_j}) \neq \text{cut}(A_k, \overline{A_k})$ if $j \neq k$, and since

$$\text{Ncut}(A_1, \dots, A_K) = \sum_{j=1}^K \frac{\text{cut}(A_j, \overline{A_j})}{\text{vol}(A_j)},$$

we would like to choose a_j, b_j so that

$$\frac{\text{cut}(A_j, \overline{A_j})}{\text{vol}(A_j)} = \frac{(X^j)^\top L X^j}{(X^j)^\top D X^j} \quad j = 1, \dots, K,$$

because this implies that

$$\begin{aligned} \mu(X) = \text{Ncut}(A_1, \dots, A_K) &= \sum_{j=1}^K \frac{\text{cut}(A_j, \overline{A_j})}{\text{vol}(A_j)} \\ &= \sum_{j=1}^K \frac{(X^j)^\top L X^j}{(X^j)^\top D X^j}. \end{aligned}$$

We find the condition

$$2\alpha_j b_j (b_j - a_j) = d b_j^2.$$

The above equation is trivially satisfied if $b_j = 0$.

If $b_j \neq 0$, then

$$a_j = \frac{2\alpha_j - d}{2\alpha_j} b_j.$$

This choice seems more complicated than the choice $b_j = 0$, so we will opt for the choice $b_j = 0$, $j = 1, \dots, K$.

With this choice, we get

$$(X^j)^\top D X^j = \alpha_j a_j^2.$$

Thus, it makes sense to pick

$$a_j = \frac{1}{\sqrt{\alpha_j}} = \frac{1}{\sqrt{\text{vol}(A_j)}}, \quad j = 1, \dots, K,$$

which is the solution presented in von Luxburg [15]. This choice also corresponds to the scaled partition matrix used in Yu [16] and Yu and Shi [17].

When $N = 10$ and $K = 4$, an example of a matrix X representing the partition of $V = \{v_1, v_2, \dots, v_{10}\}$ into the four blocks

$$\{A_1, A_2, A_3, A_4\} = \{\{v_2, v_4, v_6\}, \{v_1, v_5\}, \{v_3, v_8, v_{10}\}, \{v_7, v_9\}\},$$

is shown below:

$$X = \begin{pmatrix} 0 & a_2 & 0 & 0 \\ a_1 & 0 & 0 & 0 \\ 0 & 0 & a_3 & 0 \\ a_1 & 0 & 0 & 0 \\ 0 & a_2 & 0 & 0 \\ a_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & a_4 \\ 0 & 0 & a_3 & 0 \\ 0 & 0 & 0 & a_4 \\ 0 & 0 & a_3 & 0 \end{pmatrix}.$$

Let us now consider the problem of finding necessary and sufficient conditions for a matrix X to represent a partition of V .

When $b_j = 0$, the pairwise disjointness of the A_i is captured by the orthogonality of the X^i :

$$(X^i)^\top X^j = 0, \quad 1 \leq i, j \leq K, \quad i \neq j. \quad (*)$$

When we formulate our minimization problem in terms of Rayleigh ratios, conditions on the quantities $(X^i)^\top DX^i$ show up, and it is more convenient to express the orthogonality conditions using the quantities $(X^i)^\top DX^j$ instead of the $(X^i)^\top X^j$, because these various conditions can be combined into a single condition involving the matrix $X^\top DX$.

Now, because D is a diagonal matrix with positive entries and because the nonzero entries in each column of X have the same sign, for any $i \neq j$, the condition

$$(X^i)^\top X^j = 0$$

is equivalent to

$$(X^i)^\top DX^j = 0. \tag{**}$$

Observe that the orthogonality conditions $(*)$ (and $(**)$) are equivalent to the fact that every row of X has at most one nonzero entry.

Each A_j is nonempty iff $X^j \neq 0$, and the fact that the union of the A_j is V is captured by the fact that each row of X must have some nonzero entry (every vertex appears in some block).

It is not immediately obvious how to state conveniently this condition in matrix form.

Since every row of any matrix X representing a partition has a single nonzero entry a_j , we have

$$X^\top X = \text{diag} \left(n_1 a_1^2, \dots, n_K a_K^2 \right),$$

where n_j is the number of elements in A_j , the j th block of the partition.

Therefore, the condition for the columns of X to be nonzero is

$$\det(X^\top X) \neq 0.$$

Another condition which does not involve explicitly a determinant and is scale-invariant stems from the observation that not only

$$X^\top X = \text{diag} \left(n_1 a_1^2, \dots, n_K a_K^2 \right),$$

but

$$X^\top \mathbf{1}_N = \begin{pmatrix} n_1 a_1 \\ \vdots \\ n_K a_K \end{pmatrix},$$

and these equations imply that

$$(X^\top X)^{-1} X^\top \mathbf{1}_N = \begin{pmatrix} \frac{1}{a_1} \\ \vdots \\ \frac{1}{a_K} \end{pmatrix},$$

and thus

$$X(X^\top X)^{-1} X^\top \mathbf{1}_N = \mathbf{1}_N. \quad (\dagger)$$

Note that because the columns of X are linearly independent, $(X^\top X)^{-1} X^\top$ is the pseudo-inverse X^+ of X .

Consequently, if $X^\top X$ is invertible, condition $(\dagger)$ can also be written as

$$X X^+ \mathbf{1}_N = \mathbf{1}_N.$$

However, it is well known that XX^+ is the orthogonal projection of $\mathbb{R}^K$ onto the range of X (see Gallier [6], Section 14.1), so the condition $XX^+\mathbf{1}_N = \mathbf{1}_N$ is equivalent to the fact that $\mathbf{1}_N$ belongs to the range of X .

In retrospect, this should have been obvious since the columns of a solution X satisfy the equation

$$a_1^{-1}X^1 + \cdots + a_K^{-1}X^K = \mathbf{1}_N.$$

We emphasize that it is important to use conditions that are invariant under multiplication by a nonzero scalar, because the Rayleigh ratio is scale-invariant, and it is crucial to take advantage of this fact to make the denominators go away.

If we let

$$\mathcal{X} = \left\{ [X^1 \dots X^K] \mid X^j = a_j(x_1^j, \dots, x_N^j), x_i^j \in \{1, 0\}, \right. \\ \left. a_j \in \mathbb{R}, X^j \neq 0 \right\}$$

(note that the condition $X^j \neq 0$ implies that $a_j \neq 0$), then the set of matrices representing partitions of V into K blocks is

$$\mathcal{K} = \left\{ X = [X^1 \dots X^K] \mid X \in \mathcal{X}, \right. \\ (X^i)^\top D X^j = 0, \\ 1 \leq i, j \leq K, i \neq j, \\ \left. X(X^\top X)^{-1} X^\top \mathbf{1} = \mathbf{1} \right\}.$$

As in the case $K = 2$, to be rigorous, the solution are really K -tuples of points in $\mathbb{RP}^{N-1}$, so our solution set is really

$$\mathbb{P}(\mathcal{K}) = \left\{ (\mathbb{P}(X^1), \dots, \mathbb{P}(X^K)) \mid [X^1 \ \dots \ X^K] \in \mathcal{K} \right\}.$$

In view of the above, we have our first formulation of K -way clustering of a graph using normalized cuts, called problem PNC1 (the notation PNCX is used in Yu [16], Section 2.1):

**K -way Clustering of a graph using Normalized
Cut, Version 1:
Problem PNC1**

$$\begin{aligned}
& \text{minimize} && \sum_{j=1}^K \frac{(X^j)^\top L X^j}{(X^j)^\top D X^j} \\
& \text{subject to} && (X^i)^\top D X^j = 0, \quad 1 \leq i, j \leq K, \ i \neq j, \\
& && X(X^\top X)^{-1} X^\top \mathbf{1} = \mathbf{1}, \\
& && X \in \mathcal{X}.
\end{aligned}$$

As in the case $K = 2$, the solutions that we are seeking are K -tuples $(\mathbb{P}(X^1), \dots, \mathbb{P}(X^K))$ of points in $\mathbb{RP}^{N-1}$ determined by their homogeneous coordinates $X^1, \dots, X^K$.

Remark:

Because

$$\begin{aligned} (X^j)^\top L X^j &= (X^j)^\top D X^j - (X^j)^\top W X^j \\ &= \text{vol}(A_j) - (X^j)^\top W X^j, \end{aligned}$$

Instead of minimizing

$$\mu(X^1, \dots, X^K) = \sum_{j=1}^K \frac{(X^j)^\top L X^j}{(X^j)^\top D X^j},$$

we can maximize

$$\epsilon(X^1, \dots, X^K) = \sum_{j=1}^K \frac{(X^j)^\top W X^j}{(X^j)^\top D X^j},$$

since

$$\epsilon(X^1, \dots, X^K) = K - \mu(X^1, \dots, X^K).$$

Theoretically, minimizing $\mu(X^1, \dots, X^K)$ is equivalent to maximizing $\epsilon(X^1, \dots, X^K)$, but from a practical point of view, it is preferable to maximize $\epsilon(X^1, \dots, X^K)$.

This is because minimizing solutions of μ are obtained from (unit) eigenvectors corresponding to the K *smallest* eigenvalues of $L_{\text{sym}} = D^{-1/2}LD^{-1/2}$ (by multiplying these eigenvectors by $D^{1/2}$).

However, numerical methods for computing eigenvalues and eigenvectors of a symmetric matrix do much better at computing largest eigenvalues.

Since $L_{\text{sym}} = I - D^{-1/2}WD^{-1/2}$, the eigenvalues of L_{sym} listed in increasing order correspond to the eigenvalues of $I - L_{\text{sym}} = D^{-1/2}WD^{-1/2}$ listed in decreasing order.

Furthermore, v is an eigenvector of L_{sym} for the i th smallest eigenvalue ν_i iff v is an eigenvector of $I - L_{\text{sym}}$ for the $(N + 1 - i)$ th largest eigenvalue ν_i .

Therefore, it is preferable to find the *largest* eigenvalues of $I - L_{\text{sym}} = D^{-1/2}WD^{-1/2}$ and their eigenvectors.

In fact, since the eigenvalues of L_{sym} are in the range $[0, 2]$, the eigenvalues of $2I - L_{\text{sym}} = I + D^{-1/2}WD^{-1/2}$ are also in the range $[0, 2]$ (that is, $I + D^{-1/2}WD^{-1/2}$ is positive semidefinite).

Let us now show how our original formulation (PNC1) can be converted to a more convenient form, by chasing the denominators in the Rayleigh ratios, and by expressing the objective function in terms of the *trace* of a certain matrix.

Proposition 3.1. *For any orthogonal $K \times K$ matrix R , any symmetric $N \times N$ matrix A , and any $N \times K$ matrix $X = [X^1 \cdots X^K]$, the following properties hold:*

(1) $\mu(X) = \text{tr}(\Lambda^{-1} X^\top L X)$, where

$$\Lambda = \text{diag}((X^1)^\top D X^1, \dots, (X^K)^\top D X^K).$$

(2) If $(X^1)^\top D X^1 = \cdots = (X^K)^\top D X^K = \alpha^2$, then

$$\mu(X) = \mu(XR) = \frac{1}{\alpha^2} \text{tr}(X^\top L X).$$

(3) The condition $X^\top A X = \alpha^2 I$ is preserved if X is replaced by XR .

(4) The condition $X(X^\top X)^{-1} X^\top \mathbf{1} = \mathbf{1}$ is preserved if X is replaced by XR .

Now, by Proposition 3.1(1) and the fact that the conditions in PNC1 are scale-invariant, we are led to the following formulation of our problem:

$$\begin{aligned}
& \text{minimize} && \text{tr}(X^\top LX) \\
& \text{subject to} && (X^i)^\top DX^j = 0, \quad 1 \leq i, j \leq K, \ i \neq j, \\
& && (X^j)^\top DX^j = 1, \quad 1 \leq j \leq K, \\
& && X(X^\top X)^{-1}X^\top \mathbf{1} = \mathbf{1}, \\
& && X \in \mathcal{X}.
\end{aligned}$$

Conditions on lines 2 and 3 can be combined in the equation

$$X^\top DX = I,$$

and, we obtain the following formulation of our minimization problem:

**K -way Clustering of a graph using Normalized
Cut, Version 2:
Problem PNC2**

$$\begin{aligned}
 &\text{minimize} && \text{tr}(X^\top LX) \\
 &\text{subject to} && X^\top DX = I, \\
 &&& X(X^\top X)^{-1}X^\top \mathbf{1} = \mathbf{1}, \quad X \in \mathcal{X}.
 \end{aligned}$$

Because problem PNC2 requires the constraint $X^\top DX = I$ to be satisfied, it does not have the same set of solutions as problem PNC1.

Nevertheless, problem PNC2 is equivalent to problem PNC1, in the sense that for every minimal solution $(X^1, \dots, X^K)$ of PNC1, $((X^1)^\top D X^1)^{-1/2} X^1, \dots, ((X^K)^\top D X^K)^{-1/2} X^K$ is a minimal solution of PNC2 (with the same minimum for the objective functions), and that for every minimal solution $(Z^1, \dots, Z^K)$ of PNC2, $(\lambda_1 Z^1, \dots, \lambda_K Z^K)$ is a minimal solution of PNC1, for all $\lambda_i \neq 0$, $i = 1, \dots, K$ (with the same minimum for the objective functions).

In other words, problems PNC1 and PNC2 have the same set of minimal solutions as K -tuples of points $(\mathbb{P}(X^1), \dots, \mathbb{P}(X^K))$ in $\mathbb{RP}^{N-1}$ determined by their homogeneous coordinates $X^1, \dots, X^K$.

Formulation PNC2 reveals that finding a minimum normalized cut has a geometric interpretation in terms of the graph drawings discussed in Section 2.1.

Indeed, PNC2 has the following equivalent formulation: Find a minimal energy graph drawing X in $\mathbb{R}^K$ of the weighted graph $G = (V, W)$ such that:

1. The matrix X is orthogonal with respect to the inner product $\langle -, - \rangle_D$ in $\mathbb{R}^N$ induced by D , with

$$\langle x, y \rangle_D = x^\top D y, \quad x, y \in \mathbb{R}^N.$$

2. The rows of X are nonzero; this means that no vertex $v_i \in V$ is assigned to the origin of $\mathbb{R}^K$ (the zero vector 0_K).
3. Every vertex v_i is assigned a point of the form $(0, \dots, 0, a_j, 0, \dots, 0)$ on some axis (in $\mathbb{R}^K$).
4. Every axis in $\mathbb{R}^K$ is assigned at least some vertex.

Condition 1 can be reduced to the standard condition for graph drawings ($R^\top R = I$) by making the change of variable $Y = D^{1/2}X$ or equivalently $X = D^{-1/2}Y$. Indeed,

$$\text{tr}(X^\top LX) = \text{tr}(Y^\top D^{-1/2}LD^{-1/2}Y),$$

so we use the normalized Laplacian $L_{\text{sym}} = D^{-1/2}LD^{-1/2}$ instead of L ,

$$X^\top DX = Y^\top Y = I,$$

and conditions (2), (3), (4) are preserved under the change of variable $Y = D^{1/2}X$, since $D^{1/2}$ is invertible.

However, conditions (2), (3), (4) are “hard” constraints, especially condition (3).

In fact, condition (3) implies that the columns of X are orthogonal with respect to both the Euclidean inner product and the inner product $\langle -, - \rangle_D$, so condition (1) is redundant, except for the fact that it prescribes the norm of the columns, but this is not essential due to the projective nature of the solutions.

The main problem in finding a good relaxation of problem PNC2 is that it is very difficult to enforce the condition $X \in \mathcal{X}$.

Also, the solutions X are not preserved under arbitrary rotations, but only by very special rotations which leave $\mathcal{X}$ invariant (they exchange the axes).

The first natural relaxation of problem PNC2 is to drop the condition that $X \in \mathcal{X}$, and we obtain the

Problem $(*_2)$

$$\begin{aligned} & \text{minimize} && \text{tr}(X^\top LX) \\ & \text{subject to} && X^\top DX = I, \\ & && X(X^\top X)^{-1}X^\top \mathbf{1} = \mathbf{1}. \end{aligned}$$

Actually, since the discrete solutions $X \in \mathcal{X}$ that we are ultimately seeking are solutions of problem PNC1, the preferred relaxation is the one obtained from problem PNC1 by dropping the condition $X \in \mathcal{X}$, and simply requiring that $X^j \neq 0$, for $j = 1, \dots, K$:

Problem $(*_1)$

$$\begin{aligned}
& \text{minimize} && \sum_{j=1}^K \frac{(X^j)^\top L X^j}{(X^j)^\top D X^j} \\
& \text{subject to} && (X^i)^\top D X^j = 0, X^j \neq 0 \quad 1 \leq i, j \leq K, i \neq j, \\
& && X(X^\top X)^{-1} X^\top \mathbf{1} = \mathbf{1}.
\end{aligned}$$

Now that we dropped the condition $X \in \mathcal{X}$, it is not clear that $X^\top X$ is invertible in $(*_1)$ and $(*_2)$.

However, since the columns of X are nonzero and D -orthogonal, they must be linearly independent, so X has rank K and $X^\top X$ is invertible.

As we explained before, every solution $Z = [Z^1, \dots, Z^K]$ of problem $(*_1)$ yields a solution of problem $(*_2)$ by normalizing each Z^j by $((Z^j)^\top D Z^j)^{1/2}$, and conversely for every solution $Z = [Z^1, \dots, Z^K]$ of problem $(*_2)$, the K -tuple $[\lambda_1 Z^1, \dots, \lambda_K Z^K]$ is a solution of problem $(*_1)$, where $\lambda_j \neq 0$ for $j = 1, \dots, K$.

Furthermore, by Proposition 3.1, for every orthogonal matrix $R \in \mathbf{O}(K)$ and for every solution X of $(*_2)$, the matrix XR is also a solution of $(*_2)$.

Since Proposition 3.1(2) requires that all $(X^j)^\top D X^j$ have the same value in order to have $\mu(X) = \mu(XR)$, in general, if X is a solution of $(*_1)$, the matrix XR is not necessarily a solution of $(*_1)$.

However, every solution X of $(*_2)$ is also a solution of $(*_1)$, for every $R \in \mathbf{O}(K)$, XR is a solution of both $(*_2)$ and $(*_1)$, and since $(*_1)$ is scale-invariant, for every diagonal invertible matrix Λ , the matrix $XR\Lambda$ is a solution of $(*_1)$.

In summary, every solution Z of problem $(*_2)$ yields a *family of solutions* of problem $(*_1)$; namely, all matrices of the form $ZR\Lambda$, where $R \in \mathbf{O}(K)$ and Λ is a diagonal invertible matrix.

We will take advantage of this fact in looking for a discrete solution X “close” to a solution Z of the relaxed problem $(*_2)$.

Observe that a matrix is of the form $R\Lambda$ with $R \in \mathbf{O}(K)$ and Λ a diagonal invertible matrix iff its columns are nonzero and pairwise orthogonal.

Recall that if $X^\top X$ is invertible (which is the case), condition $X(X^\top X)^{-1}X^\top \mathbf{1} = \mathbf{1}$ is equivalent to $XX^+ \mathbf{1} = \mathbf{1}$, which is also equivalent to the fact that $\mathbf{1}$ is in the range of X .

If we make the change of variable $Y = D^{1/2}X$ or equivalently $X = D^{-1/2}Y$, the condition that $\mathbf{1}$ is in the range of X becomes the condition that $D^{1/2}\mathbf{1}$ is in the range of Y , which is equivalent to

$$YY^+ D^{1/2} \mathbf{1} = D^{1/2} \mathbf{1}.$$

However, since $Y^\top Y = I$, we have

$$Y^+ = Y^\top,$$

so we get the equivalent problem

Problem (**₂)

$$\begin{array}{ll}
\text{minimize} & \text{tr}(Y^\top D^{-1/2} L D^{-1/2} Y) \\
\text{subject to} & Y^\top Y = I, \\
& Y Y^\top D^{1/2} \mathbf{1} = D^{1/2} \mathbf{1}.
\end{array}$$

We pass from a solution Y of problem (**₂) to a solution Z of problem (*₂) by $Z = D^{-1/2} Y$.

It is not a priori obvious that the minimum of $\text{tr}(Y^\top L_{\text{sym}} Y)$ over all $N \times K$ matrices Y satisfying $Y^\top Y = I$ is equal to the sum $\nu_1 + \cdots + \nu_K$ of the first K eigenvalues of $L_{\text{sym}} = D^{-1/2} L D^{-1/2}$.

Fortunately, the Poincaré separation theorem (Proposition A.3) guarantees that the sum of the K smallest eigenvalues of L_{sym} is a lower bound for $\text{tr}(Y^\top L_{\text{sym}} Y)$.

Furthermore, if we temporarily ignore the second constraint, the minimum of problem $(**_2)$ is achieved by any K unit eigenvectors $(u_1, \dots, u_K)$ associated with the smallest eigenvalues

$$0 = \nu_1 \leq \nu_2 \leq \dots \leq \nu_K$$

of L_{sym} .

We may assume that $\nu_2 > 0$, namely that the underlying graph is connected (otherwise, we work with each connected component), in which case $Y^1 = D^{1/2} \mathbf{1} / \|D^{1/2} \mathbf{1}\|_2$, because $\mathbf{1}$ is in the nullspace of L .

Since $Y^1 = D^{1/2}\mathbf{1} / \|D^{1/2}\mathbf{1}\|_2$, the vector $D^{1/2}\mathbf{1}$ is in the range of Y , so the condition

$$YY^\top D^{1/2}\mathbf{1} = D^{1/2}\mathbf{1}$$

is also satisfied. Then, $Z = D^{-1/2}Y$ with $Y = [u_1 \dots u_K]$ yields a minimum of our relaxed problem $(*_2)$ (the second constraint is satisfied because $\mathbf{1}$ is in the range of Z).

By Proposition 1.6, the vectors Z^j are eigenvectors of L_{rw} associated with the eigenvalues $0 = \nu_1 \leq \nu_2 \leq \dots \leq \nu_K$.

Recall that $\mathbf{1}$ is an eigenvector for the eigenvalue $\nu_1 = 0$, and $Z^1 = \mathbf{1} / \|D^{1/2}\mathbf{1}\|_2$.

Because, $(Y^i)^\top Y^j = 0$ whenever $i \neq j$, we have

$$(Z^i)^\top D Z^j = 0, \quad \text{whenever } i \neq j.$$

This implies that $Z^2, \dots, Z^K$ are all orthogonal to $D\mathbf{1}$, and thus, that each Z^j has both some positive and some negative coordinate, for $j = 2, \dots, K$.

The conditions $(Z^i)^\top D Z^j = 0$ do not necessarily imply that Z^i and Z^j are orthogonal (w.r.t. the Euclidean inner product), but we can obtain a solution of Problems $(*_2)$ and $(*_1)$ achieving the same minimum for which distinct columns Z^i and Z^j are simultaneously orthogonal and D -orthogonal, by multiplying Z by some $K \times K$ orthogonal matrix R on the right.

Indeed, if Z is a solution of $(*_2)$ obtained as above, the $K \times K$ symmetric matrix $Z^\top Z$ can be diagonalized by some orthogonal $K \times K$ matrix R as

$$Z^\top Z = R\Sigma R^\top,$$

where Σ is a diagonal matrix, and thus,

$$R^\top Z^\top Z R = (ZR)^\top ZR = \Sigma,$$

which shows that the columns of ZR are orthogonal.

By Proposition 3.1, ZR also satisfies the constraints of $(*_2)$ and $(*_1)$, and $\text{tr}((ZR)^\top L(ZR)) = \text{tr}(Z^\top LZ)$.

Remark: Since Y has linearly independent columns (in fact, orthogonal) and since $Z = D^{-1/2}Y$, the matrix Z also has linearly independent columns, so $Z^\top Z$ is positive definite and the entries in Σ are all positive.

Also, instead of computing $Z^\top Z$ explicitly and diagonalizing it, the matrix R can be found by computing an SVD of Z .

3.4 Finding a Discrete Solution Close to a Continuous Approximation

The next step is to find an exact solution

$(\mathbb{P}(X^1), \dots, \mathbb{P}(X^K)) \in \mathbb{P}(\mathcal{K})$ which is the closest (in a suitable sense) to our approximate solution $(Z^1, \dots, Z^K)$.

Recall from Section 3.3 that every solution Z of problem $(*_2)$ yields a *family of solutions* of problem $(*_1)$; namely, all matrices of the form ZQ , where Q is a $K \times K$ matrix with nonzero and pairwise orthogonal columns.

Since the solutions ZQ of $(*_1)$ are all equivalent (they yield the same minimum for the normalized cut), it makes sense to look for a discrete solution X closest to one of these ZQ .

We view $\mathcal{K}$ as a subset of $(\mathbb{RP}^{N-1})^K$.

Because $\mathcal{K}$ is closed under the antipodal map, it can be shown that for every j ($1 \leq j \leq K$), minimizing the distance $d(\mathbb{P}(X^j), \mathbb{P}(Z^j))$ on $\mathbb{RP}^{N-1}$ is equivalent to minimizing $\|X^j - Z^j\|_2$, where X^j and Z^j are representatives of $\mathbb{P}(X^j)$ and $\mathbb{P}(Z^j)$ on the unit sphere (if we use the Riemannian metric on $\mathbb{RP}^{N-1}$ induced by the Euclidean metric on $\mathbb{R}^N$).

Then, if we use the product distance on $(\mathbb{RP}^{N-1})^K$ given by

$$\begin{aligned} d((\mathbb{P}(X^1), \dots, \mathbb{P}(X^K)), (\mathbb{P}(Z^1), \dots, \mathbb{P}(Z^K))) \\ = \sum_{j=1}^K d(\mathbb{P}(X^j), \mathbb{P}(Z^j)), \end{aligned}$$

minimizing the distance

$d((\mathbb{P}(X^1), \dots, \mathbb{P}(X^K)), (\mathbb{P}(Z^1), \dots, \mathbb{P}(Z^K)))$ in $(\mathbb{RP}^{N-1})^K$ is equivalent to minimizing

$$\sum_{j=1}^K \|X^j - Z^j\|_2, \quad \text{subject to}$$

$$\|X^j\|_2 = \|Z^j\|_2 \quad (j = 1, \dots, K).$$

We are not aware of any optimization method to solve the above problem, which seems difficult to tackle due to constraints $\|X^j\|_2 = \|Z^j\|_2$ ($j = 1, \dots, K$).

Therefore, we drop these constraints and attempt to minimize

$$\|X - Z\|_F^2 = \sum_{j=1}^K \|X^j - Z^j\|_2^2,$$

the Frobenius norm of $X - Z$. This is implicitly the choice made by Yu.

Inspired by Yu [16] and the previous discussion, given a solution Z of problem $(*_2)$, we look for pairs (X, Q) with $X \in \mathcal{X}$ and where Q is a $K \times K$ matrix with nonzero and pairwise orthogonal columns, with $\|X\|_F = \|Z\|_F$, that minimize

$$\varphi(X, Q) = \|X - ZQ\|_F.$$

Yu [16] and Yu and Shi [17] consider the special case where $Q \in \mathbf{O}(K)$.

We consider the more general case where $Q = R\Lambda$, with $R \in \mathbf{O}(K)$ and Λ is a diagonal invertible matrix.

The key to minimizing $\|X - ZQ\|_F$ rests on the following result:

$$\|X - ZQ\|_F^2 = \|X\|_F^2 - 2\text{tr}(Q^\top Z^\top X) + \text{tr}(Z^\top ZQQ^\top).$$

Therefore, since $\|X\|_F = \|Z\|_F$ is fixed, minimizing $\|X - ZQ\|_F^2$ is equivalent to
 minimizing $-2\text{tr}(Q^\top Z^\top X) + \text{tr}(Z^\top ZQQ^\top)$.

This is a hard problem because it is a nonlinear optimization problem involving two matrix unknowns X and Q .

To simplify the problem, we proceed by *alternating steps* during which we *minimize* $\varphi(X, Q) = \|X - ZQ\|_F$ *with respect to* X *holding* Q *fixed*, and steps during which we *minimize* $\varphi(X, Q) = \|X - ZQ\|_F$ *with respect to* Q *holding* X *fixed*.

This *second step in which* X *is held fixed* has been studied, but it is still a hard problem for which no closed-form solution is known.

Consequently, we further simplify the problem.

Since Q is of the form $Q = R\Lambda$ where $R \in \mathbf{O}(K)$ and Λ is a diagonal invertible matrix, we minimize $\|X - ZR\Lambda\|_F$ in two stages.

1. We set $\Lambda = I$ and find $R \in \mathbf{O}(K)$ that minimizes $\|X - ZR\|_F$.
2. Given X , Z , and R , find a diagonal invertible matrix Λ that minimizes $\|X - ZR\Lambda\|_F$.

The matrix $R\Lambda$ is not a minimizer of $\|X - ZR\Lambda\|_F$ in general, but it is an improvement on R alone, and both stages can be solved quite easily.

In *stage 1*, the matrix $Q = R$ is orthogonal, so $QQ^\top = I$, and since Z and X are given, the problem reduces to minimizing $-2\text{tr}(Q^\top Z^\top X)$; that is, *maximizing* $\text{tr}(Q^\top Z^\top X)$.

To solve this problem, we need the following proposition.

Proposition 3.2. *For any $n \times n$ matrix A and any orthogonal matrix Q , we have*

$$\max\{\text{tr}(QA) \mid Q \in \mathbf{O}(n)\} = \sigma_1 + \cdots + \sigma_n,$$

where $\sigma_1 \geq \cdots \geq \sigma_n$ are the singular values of A . Furthermore, this maximum is achieved by $Q = VU^\top$, where $A = U\Sigma V^\top$ is any SVD for A .

As a corollary of Proposition 3.2 (with $A = Z^\top X$ and $Q = R^\top$), we get the following result (see Golub and Van Loan [8], Section 12.4.1):

Proposition 3.3. *For any two fixed $N \times K$ matrices X and Z , the minimum of the set*

$$\{\|X - ZR\|_F \mid R \in \mathbf{O}(K)\}$$

is achieved by $R = UV^\top$, for any SVD decomposition $U\Sigma V^\top = Z^\top X$ of $Z^\top X$.

The following proposition takes care of *stage 2*.

Proposition 3.4. *For any two fixed $N \times K$ matrices X and Z , where Z has no zero column, there is a unique diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_K)$ minimizing $\|X - Z\Lambda\|_F$ given by*

$$\lambda_j = \frac{(Z^\top X)_{jj}}{\|Z^j\|_2^2} \quad j = 1, \dots, K.$$

It should be noted that Proposition 3.4 does not guarantee that Λ is invertible. For example, for

$$X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & -1 \end{pmatrix},$$

we have

$$Z^\top X = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix},$$

so $\lambda_2 = 0$. When Proposition 3.4 yields a singular matrix, we skip stage 2 (we set $\Lambda = I$).

We now deal with *step 1, where $Q = R\Lambda$ is held fixed.*

For fixed Z and Q , we would like to find some $X \in \mathcal{K}$ with $\|X\|_F = \|Z\|_F$ so that $\|X - ZQ\|_F$ is minimal.

Without loss of generality, we may assume that the entries $a_1, \dots, a_K$ occurring in the matrix X are positive and all equal to some common value $a \neq 0$.

Recall that a matrix $X \in \mathcal{X}$ has the property that every row contains exactly one nonzero entry, and that every column is nonzero.

To find $X \in \mathcal{K}$, first we find the shape $\hat{X}$ of X , which is the matrix obtained from X by rescaling the columns of X so that $\hat{X}$ has entries $+1, 0$.

The problem is to decide for each row, which column contains the nonzero entry.

After having found $\widehat{X}$, we rescale its columns so that $\|X\|_F = \|Z\|_F$.

Since

$$\|X - ZQ\|_F^2 = \|X\|_F^2 - 2\text{tr}(Q^\top Z^\top X) + \text{tr}(Z^\top ZQQ^\top),$$

minimizing $\|X - ZQ\|_F$ is equivalent to *maximizing*

$$\text{tr}(Q^\top Z^\top X) = \text{tr}((ZQ)^\top X) = \text{tr}(X(ZQ)^\top),$$

and since the i th row of X contains a single nonzero entry a in column j_i ($1 \leq j_i \leq K$), if we write $Y = ZQ$, then

$$\text{tr}(XY^\top) = a \sum_{i=1}^N y_{ij_i}. \quad (*)$$

By (*), since $a > 0$, the quantity $\text{tr}(XY^\top)$ is maximized iff y_{ij_i} is maximized for $i = 1, \dots, N$; *this is achieved if for the i th row of X , we pick a column index ℓ such that $y_{i\ell}$ is maximum.*

To find the shape $\hat{X}$ of X , we first find a matrix $\bar{X}$ by choosing a single nonzero entry $\bar{x}_{ij} = 1$ on row i in such a way that y_{ij} is maximum according to the following method.

If we let

$$\begin{aligned}\mu_i &= \max_{1 \leq j \leq K} y_{ij} \\ J_i &= \{j \in \{1, \dots, K\} \mid y_{ij} = \mu_i\},\end{aligned}$$

for $i = 1, \dots, N$, then

$$\bar{x}_{ij} = \begin{cases} +1 & \text{for some chosen } j \in J_i, \\ 0 & \text{otherwise.} \end{cases}$$

Of course, a single column index is chosen for each row. In our implementation, we pick the smallest index in J_i .

Unfortunately, the matrix $\overline{X}$ may not be a correct solution, because the above prescription does not guarantee that every column of $\overline{X}$ is nonzero.

When this happens, we *reassign certain nonzero entries in columns having “many” nonzero entries to zero columns, so that we get a matrix in $\mathcal{K}$.*

Suppose column j is zero. Then, we pick the leftmost index k of a column with a maximum number of 1, and if i the smallest index for which $\overline{X}_{ik} = 1$, then we set $\overline{X}_{ik} = 0$ and $\overline{X}_{ij} = 1$.

We repeat this reallocation scheme until every column is nonzero.

We obtain a new matrix $\widehat{X}$ in $\mathcal{X}$, and finally we normalize $\widehat{X}$ to obtain X , so that $\|X\|_F = \|Z\|_F$.

The above method is essentially the method described in Yu [16] and Yu and Shi [17].

The fact that $\overline{X}$ may have zero columns is not addressed by Yu.

Furthermore, it is important to make sure that X has the same norm as Z , but this normalization step is not performed in the above works.

On the other hand, the rows of Z are normalized and the resulting matrix may no longer be a correct solution of the relaxed problem.

In practice, it appears to be a good approximation of a discrete solution; see option (3) of the initialization methods for Z described below.

We found that the following heuristic is quite helpful in finding a better discrete solution X .

Any matrix obtained by flipping the signs of some of the columns of a solution ZR of problem $(*_2)$ is still a solution.

Moreover, all entries in X are nonnegative. It follows that a “good” solution ZQ (that is, close to a discrete solution) should have the property that the average of each of its column is nonnegative.

Therefore, given a solution ZQ of problem $(*_2)$, we compute ZQ_p , defined such that if the average of column $(ZQ)^j$ is negative, then $(ZQ_p)^j = -(ZQ)^j$, else $(ZQ_p)^j = (ZQ)^j$.

It follows that the average of every column in ZQp is nonnegative. Then, we apply the above procedure to find discrete solutions X and X_p closest to ZQ and ZQ_p respectively, and we pick the solution corresponding to $\min\{\|X - ZQ\|_F, \|X_p - ZQ_p\|_F\}$.

Flipping signs of columns of ZQ correspond to a diagonal matrix R_p with entries ± 1 , a very special kind of orthogonal matrix.

In summary, the procedure for finding a discrete X close to a continuous ZQ also updates Q to QR_p .

This step appears to be very effective for finding a good initial X .

The method due to Yu and Shi (see Yu [16] and Yu and Shi [17]) to find $X \in \mathcal{K}$ and $Q = R\Lambda$ with $R \in \mathbf{O}(K)$ and Λ diagonal invertible that minimize

$\varphi(X, Q) = \|X - ZQ\|_F$ is to *alternate steps* during which *either Q is held fixed* (step PODX) *or X is held fixed* (step PODR), except that Yu and Shi consider the special case where $\Lambda = I$.

- (1) In step PODX, the next discrete solution X^* is obtained from the previous pair (Q^*, Z) by computing $\bar{X}$ and then $X^* = \hat{X}$ from $Y = ZQ^*$, as just explained above.
- (2) In step PODR, the next matrix $Q^* = R\Lambda$ is obtained from the previous pair (X^*, Z) by first computing

$$R = UV^\top,$$

for any SVD decomposition $U\Sigma V^\top$ of $Z^\top X^*$, and then computing Λ from X^* and ZR using Proposition 3.4. If Λ is singular, then set $\Lambda = I$.

We keep track of the progress of the procedure by computing $\varphi(X^*, Q^*) = \|X^* - ZQ^*\|_F$ after every step and checking that X^* or $\varphi(X^*, Q^*)$ stops changing, whichever comes first.

We observed that after a small number of steps, up to machine precision, $\varphi(X^*, Q^*)$ stops decreasing, and when this occurs the procedure halts (we also set a maximum number of steps in case $\varphi(X^*, Q^*)$ decreases for a very long time).

Moreover, looking for $Q = R\Lambda$ where $R \in \mathbf{O}(K)$ and Λ is obtained using the method of Proposition 3.4 speeds up the convergence and yields a better discrete solution X .

The process of searching for X and Q has an illuminating geometric interpretation in terms of graph drawings.

We may assume that the entries in the discrete solution X are 0 or 1.

Then the rows of the discrete solutions X correspond to the tips of the unit vectors along the coordinate axes in $\mathbb{R}^K$.

Every axis contains at least such a point, and the multiplicity of the point along the j th axis is the number of nodes in the j th block of the partition.

Similarly, the rows of Z are the nodes of a graph drawing of the weighted graph (V, W) .

Multiplying Z on the right by a $K \times K$ matrix Q (obtaining ZQ) is equivalent to multiplying $Z^\top$ on the left by $Q^\top$ (obtaining $Q^\top Z^\top$).

This means that the points in $\mathbb{R}^K$ representing the rows of ZQ are obtained by applying the linear transformation $Q^\top$ to the columns of $Z^\top$.

Thus, ZR amounts to applying the rigid motion $R^\top$ to the graph drawing Z , and $Z\Lambda$ (where Λ is a diagonal invertible matrix) amounts to stretching or shrinking the graph drawing Z in the directions of the axes.

Then, in step 2 (PODR), we are trying to deform the graph drawing given by Z using a linear map $(R\Lambda)^\top$, so that the deformed graph drawing $ZR\Lambda$ is as close as possible to X (in the sense that $\|X - ZR\Lambda\|_F$ is minimized).

In step 1 (PODX), we are trying to approximate the deformed graph drawing $ZR\Lambda$ by a discrete graph drawing X (whose nodes are the tips of the unit vectors), so that $\|X - ZR\Lambda\|_F$ is minimized.

If we are willing to give up the requirement that the deformed Z is still a solution of problem $(*_1)$, we have quite a bit of freedom in step 2.

For example, we may allow normalizing the rows.

This seems reasonable to obtain an initial transformation Q .

However, we feel uncomfortable in allowing intermediate deformed Z that are not solutions of $(*_1)$ during the iteration process.

This point should be investigated further.

In some sense, we have less freedom in step 1, since the i th row of $ZR\Lambda$ is assigned to the j th unit vector iff the index of the leftmost largest coordinate of this row is j .

If some axis has not been assigned any row of R , then we reallocate one of the points on an axis with a maximum number of points.

Figure 3.5 shows a graph (on the left) and the graph drawings X and $Z * R$ obtained by applying our method for three clusters.

The rows of X are represented by the red points along the axes, and the rows of $Z * R$ by the green points (on the right).

The original vertices corresponding to the rows of Z are represented in blue.

We can see how the two red points correspond to an edge, the three red points correspond to a triangle, and the four red points to a quadrangle.

These constitute the clusters.

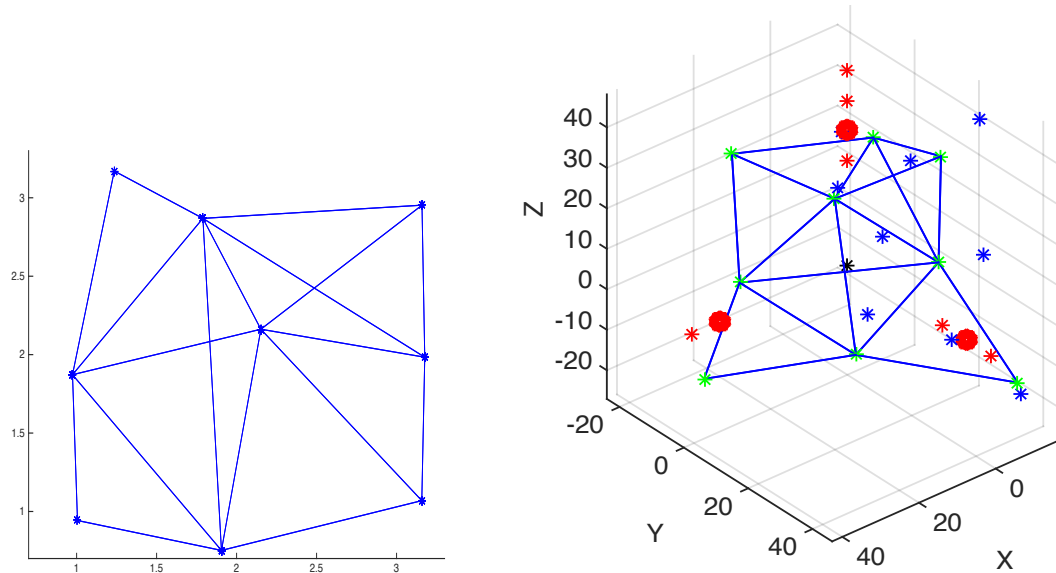


Figure 3.5: A graph and its drawing to find 3 clusters.

It remains to initialize Q^* to start the process, and then steps (1) and (2) are iterated, starting with step (1).

Actually, what we really need is a “good” initial X^* , but to find it, we need an initial R^* .

Method 1. One method is to use an orthogonal matrix denoted R_1 , such that distinct columns of ZR_1 are *simultaneously orthogonal and D-orthogonal*.

The matrix R_1 can be found by diagonalizing $Z^\top Z$ as $Z^\top Z = R_1 \Sigma R_1^\top$, as we explained at the end of Section 3.3. We write $Z_2 = ZR_1$.

Method 2. The method advocated by Yu [16] is to *pick K rows of Z that are as orthogonal to each other as possible* and to make a matrix R whose columns consist of these rows normalized to have unit length.

The intuition behind this method is that if a continuous solution Z can be sent close to a discrete solution X by a rigid motion, then many rows of Z viewed as vectors in $\mathbb{R}^K$ should be nearly orthogonal.

This way, ZR should contain at least K rows well aligned with the canonical basis vectors, and these rows are good candidates for some of the rows of the discrete solution X .

The algorithm given in Yu [16] needs a small correction, because rows are not removed from Z when they are added to R , which may cause the same row to be added several times to R .

Given the $N \times K$ matrix Z (whose columns all have the same norm), we compute a matrix R whose columns are certain rows of Z .

We use a vector $c \in \mathbb{R}^N$ to keep track of the inner products of all rows of Z with the columns $R^1, \dots, R^{k-1}$ that have been constructed so far, and initially when $k = 1$, we set $c = 0$.

The first column R^1 of R is any chosen row of Z .

Next, for $k = 2, \dots, K$, we compute all the inner products of R^{k-1} with all rows in Z , which are recorded in the vector ZR^{k-1} , and we update c as follows:

$$c = c + \mathbf{abs}(ZR^{k-1}).$$

We take the absolute values of the entries in ZR^{k-1} so that the i th entry in c is a score of how orthogonal is the i th row of Z to $R^1, \dots, R^{k-1}$.

Then, we choose R^k as any row Z_i of Z for which c_i is minimal (the customary (and ambiguous) $i = \arg \min c$), and we delete this row from Z . The process is repeated (with the updated Z) until $k = K$.

At the end of the above process, we normalize the columns of R , to obtain a matrix that we denote R_2 .

After some experimentation, we found that to obtain a better initial X^* , it is may desirable to start from a variant of the continuous solution Z obtained by solving problem $(*_2)$.

We have implemented three methods.

1. We attempt to rescale the columns of Z by some diagonal invertible matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_K)$, so that the rows of $Z\Lambda$ sum to 1 as much as possible in the least-squares sense.

Since the vector of sums of rows of $Z\Lambda$ is $Z\Lambda\mathbf{1}_K = Z\lambda$, with $\lambda^\top = (\lambda_1, \dots, \lambda_K)$, the least-squares problem is to minimize

$$\|Z\lambda - \mathbf{1}_N\|_2^2,$$

and since Z has rank K , the solution is $\lambda = (Z^\top Z)^{-1}Z^\top \mathbf{1}_N$, and thus,

$$\Lambda = \text{diag}((Z^\top Z)^{-1}Z^\top \mathbf{1}_N).$$

The matrix Λ is singular if some of the columns of Z sum to 0.

This happens for regular graphs, where the degree matrix is a multiple of the identity.

There are also cases where some of the λ_j are very small, so we use a tolerance factor to prevent this, and in case of failure, we set $\Lambda = I$.

In case of failure, we may also use ZR_1 instead of Z , where R_1 is the orthogonal matrix that makes ZR_1 both D -orthogonal and orthogonal.

2. We attempt to rescale the columns of Z by some diagonal invertible matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_K)$, so that the rows of $Z\Lambda$ have unit length as much as possible in the least-squares sense.

Since the square-norm of the i th row of $Z\Lambda$ is

$$\sum_{j=1}^K z_{ij}^2 \lambda_j^2,$$

if we write $Z \circ Z$ for the matrix (z_{ij}^2) of square entries of elements in Z (the Hadamard product of Z with itself), the least-squares problem is to minimize

$$\|Z \circ Z \lambda^2 - \mathbf{1}_N\|_2^2,$$

where $(\lambda^2)^\top = (\lambda_1^2, \dots, \lambda_K^2)$.

The matrix $Z \circ Z$ may not have rank K , so the least-squares solution for λ^2 is given by the pseudo-inverse of $Z \circ Z$, as

$$\lambda^2 = (Z \circ Z)^+ \mathbf{1}_N.$$

There is no guarantee that the vector on the right-hand side has all positive entries, so the method may fail.

It may also fail when some of the λ_j are very small.

We use a tolerance factor to prevent this, and in case of failure, we set $\Lambda = I$.

3. We use a method more drastic than (2), which consists in normalizing the rows of Z . Thus, we form the matrix

$$NZ = \text{diag}((ZZ^\top)_{11}^{-1/2}, \dots, (ZZ^\top)_{NN}^{-1/2}),$$

and we return $NZ * Z$.

Unlike the methods used in (1) and (2), this method does not guarantee that $NZ * Z$ is a solution of problem $(*_1)$.

However, since the rows of Z can be interpreted as vectors in $\mathbb{R}^K$ that should align as much as possible with the canonical basis vectors of $\mathbb{R}^K$, this method makes sense as a way to come closer to a discrete solution.

In fact, we found that it does well in most cases.

We implemented a computer program that prompts the user for various options.

To avoid confusion, let us denote the original solution of problem $(*_2)$ by Z_1 , and let $Z_2 = Z_1 R_1$, as obtained by initialization method 1.

The four options are:

1. Use the original solution Z_1 of problem $(*_2)$, as well as Z_2 .
2. Apply method 1 to Z_1 and Z_2 .
3. Apply method 2 to Z_1 and Z_2 .
4. Apply method 3 to Z_1 and Z_2 .

Then, for each of these options, if we denote by $Zinit_1$ and $Zinit_2$ the solutions returned by the method, our program computes initial solutions X_1, X_2, X_3, X_4 as follows:

1. Use $Zinit_1$ and $R = I$.
2. Use $Zinit_1$ and $R = R2a$, the matrix given by initialization method 2.
3. Use $Zinit_2$ and $R = I$.
4. Use $Zinit_2$ and $R = R2b$, the matrix given by initialization method 2.

After this, the program picks the discrete solution $X = X_i$ which corresponds to the minimum of

$$\begin{aligned} &\|X1 - Zinit1\|, \|X2 - Zinit1 * R2a\|, \\ &\|X3 - Zinit2\|, \|X4 - Zinit2 * R2b\|. \end{aligned}$$

Our experience is that options (3) and (4) tend to give better results.

However, it is harder to say whether any of the X_i does a better job than the others, although (2) and (4) seem to do slightly better than (1) and (3).

We also give the user the option in step PODR to only compute R and set $\Lambda = I$.

It appears that the more general method is hardly more expansive (because finding Λ is cheap) and always gives better results.

We also found that we obtain better results if we rescale Z (and X) so that $\|Z\|_F = 100$.

If we apply the method (using method 3 to find the initial R) to the graph associated with the the matrix W_1 shown in Figure 3.6 for $K = 4$ clusters, the algorithm converges in 3 steps and we find the clusters shown in Figure 3.7.

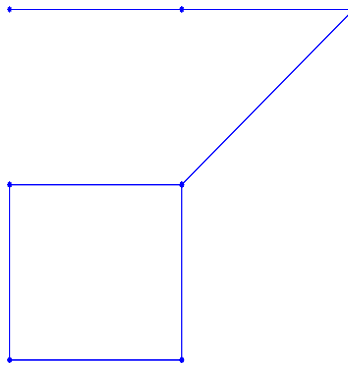


Figure 3.6: Underlying graph of the matrix W_1 .

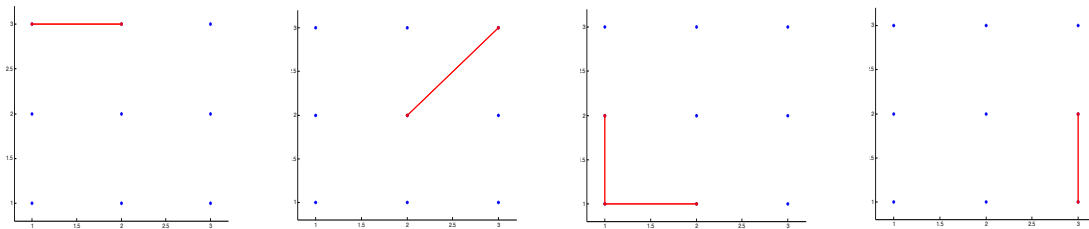


Figure 3.7: Four blocks of a normalized cut for the graph associated with W_1 .

The solution Z of the relaxed problem is

$$Z = \begin{pmatrix} -21.3146 & -0.0000 & 19.4684 & -15.4303 \\ -4.1289 & 0.0000 & 16.7503 & -15.4303 \\ -21.3146 & 32.7327 & -19.4684 & -15.4303 \\ -4.1289 & -0.0000 & 16.7503 & -15.4303 \\ 19.7150 & 0.0000 & 9.3547 & -15.4303 \\ -4.1289 & 23.1455 & -16.7503 & -15.4303 \\ -21.3146 & -32.7327 & -19.4684 & -15.4303 \\ -4.1289 & -23.1455 & -16.7503 & -15.4303 \\ 19.7150 & -0.0000 & -9.3547 & -15.4303 \end{pmatrix}.$$

We find the following sequence for $Q, Z * Q, X$:

$$Q = \begin{pmatrix} 0 & 0.6109 & -0.3446 & -0.7128 \\ -1.0000 & 0.0000 & 0.0000 & -0.0000 \\ 0.0000 & 0.5724 & 0.8142 & 0.0969 \\ -0.0000 & 0.5470 & -0.4672 & 0.6947 \end{pmatrix},$$

which is the initial Q obtained by method 1;

$$Z * Q = \begin{pmatrix} 0.0000 & -10.3162 & 30.4065 & 6.3600 \\ 0.0000 & -1.3742 & 22.2703 & -6.1531 \\ -32.7327 & -32.6044 & -1.2967 & 2.5884 \\ 0.0000 & -1.3742 & 22.2703 & -6.1531 \\ 0.0000 & 8.9576 & 8.0309 & -23.8653 \\ -23.1455 & -20.5505 & -5.0065 & -9.3982 \\ 32.7327 & -32.6044 & -1.2967 & 2.5884 \\ 23.1455 & -20.5505 & -5.0065 & -9.3982 \\ -0.0000 & -1.7520 & -7.2027 & -25.6776 \end{pmatrix}$$

$$X = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} ;$$

$$Q = \begin{pmatrix} -0.0803 & 0.8633 & -0.4518 & -0.2102 \\ -0.6485 & 0.1929 & 0.1482 & 0.7213 \\ -0.5424 & 0.0876 & 0.5546 & -0.6250 \\ -0.5281 & -0.4581 & -0.6829 & -0.2119 \end{pmatrix}$$

$$Z * Q = \begin{pmatrix} -0.6994 & -9.6267 & 30.9638 & -4.4169 \\ -0.6051 & 4.9713 & 21.6922 & -6.3311 \\ -0.8081 & -6.7218 & 14.2223 & 43.5287 \\ -0.6051 & 4.9713 & 21.6922 & -6.3311 \\ 1.4913 & 24.9075 & 6.8186 & -6.7218 \\ 2.5548 & 6.5028 & 6.5445 & 31.3015 \\ 41.6456 & -19.3507 & 4.5190 & -3.6915 \\ 32.5742 & -2.4272 & -0.3168 & -2.0882 \\ 11.6387 & 23.2692 & -3.5570 & 4.9716 \end{pmatrix}$$

$$X = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} ;$$

$$Q = \begin{pmatrix} -0.3201 & 0.7992 & -0.3953 & -0.3201 \\ -0.7071 & -0.0000 & 0.0000 & 0.7071 \\ -0.4914 & -0.0385 & 0.7181 & -0.4914 \\ -0.3951 & -0.5998 & -0.5728 & -0.3951 \end{pmatrix}$$

$$Z * Q = \begin{pmatrix} 3.3532 & -8.5296 & 31.2440 & 3.3532 \\ -0.8129 & 5.3103 & 22.4987 & -0.8129 \\ -0.6599 & -7.0310 & 3.2844 & 45.6311 \\ -0.8129 & 5.3103 & 22.4987 & -0.8129 \\ -4.8123 & 24.6517 & 7.7629 & -4.8123 \\ -0.7181 & 6.5997 & -1.5571 & 32.0146 \\ 45.6311 & -7.0310 & 3.2844 & -0.6599 \\ 32.0146 & 6.5997 & -1.5571 & -0.7181 \\ 4.3810 & 25.3718 & -5.6719 & 4.3810 \end{pmatrix}$$

$$X = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

During the next round, the exact same matrices are obtained and the algorithm stops.

Comparison of the matrices $Z * Q$ and X makes it clear that X is obtained from $Z * Q$ by retaining on every row the leftmost largest value and setting the others to 0 (non-maximum suppression).

In this example, the columns of all X were nonzero, but this may happen, for example when we apply the algorithm to the graph of Figure 3.6 to find $K = 5$ clusters shown in Figure 3.8.

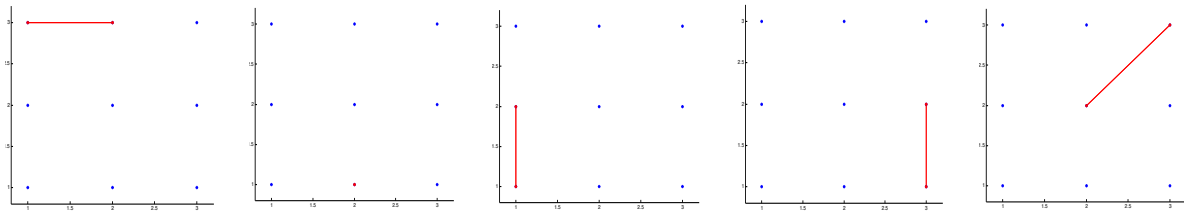


Figure 3.8: Five blocks of a normalized cut for the graph associated with W_1 .

We find that the initial value for $Z * Q$ is

$$Z * Q = \begin{pmatrix} -5.7716 & -27.5934 & 0.0000 & -9.3618 & -0.0000 \\ 5.5839 & -20.2099 & -29.7044 & -1.2471 & -0.0000 \\ -2.3489 & 1.1767 & -0.0000 & -29.5880 & -29.7044 \\ 5.5839 & -20.2099 & 29.7044 & -1.2471 & 0.0000 \\ 21.6574 & -7.2879 & 0.0000 & 8.1289 & 0.0000 \\ 8.5287 & 4.5433 & -0.0000 & -18.6493 & -21.0042 \\ -2.3489 & 1.1767 & -0.0000 & -29.5880 & 29.7044 \\ 8.5287 & 4.5433 & -0.0000 & -18.6493 & 21.0042 \\ 23.3020 & 6.5363 & -0.0000 & -1.5900 & -0.0000 \end{pmatrix}$$

The matrix $X1$ given by the above method in which we pick the leftmost largest entry on every row has a fourth row equal to 0.

The matrix $X1$ is repaired by migrating a 1 from the second entry of the first column, which contains the largest number of 1's, yielding the matrix $X2$; see below.

$$X1 = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ \textcolor{red}{1} & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix} \quad X2 = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \textcolor{blue}{1} & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Chapter 4

Signed Graphs

4.1 Signed Graphs and Signed Laplacians

Intuitively, in a weighted graph, an edge with a positive weight denotes similarity or proximity of its endpoints.

For many reasons, it is desirable to allow edges labeled with *negative weights*, the intuition being that *a negative weight indicates dissimilarity or distance*.

Weighted graphs for which the weight matrix is a symmetric matrix in which negative and positive entries are allowed are called *signed graphs*.

Such graphs (with weights $(-1, 0, +1)$) were introduced as early as 1953 by Harary [9], to model social relations involving disliking, indifference, and liking.

The problem of clustering the nodes of a signed graph arises naturally as a generalization of the clustering problem for weighted graphs.

From our perspective, we would like to know whether clustering using normalized cuts can be extended to signed graphs.

Given a signed graph $G = (V, W)$ (where W is a symmetric matrix with zero diagonal entries), the *underlying graph* of G is the graph with node set V and set of (undirected) edges $E = \{\{v_i, v_j\} \mid w_{ij} \neq 0\}$.

The first obstacle is that the degree matrix may now contain negative entries.

As a consequence, the Laplacian L may no longer be positive semidefinite, and worse, $D^{-1/2}$ may not exist.

A simple remedy is to use the absolute values of the weights in the degree matrix!

This idea applied to signed graph with weights $(-1, 0, 1)$ occurs in Hou [10]. Kolluri, Shewchuk and O'Brien [11] take the natural step of using absolute values of weights in the degree matrix in their original work on surface reconstruction from noisy point clouds.

Kunegis et al. [12] appear to be the first to make a systematic study of spectral methods applied to signed graphs.

In fact, many results in this section originate from Kunegis et al. [12].

However, it should be noted that only 2-clustering is considered in the above papers.

As we will see, the trick of using absolute values of weights in the degree matrix allows the whole machinery that we have presented to be used to attack the problem of clustering signed graphs using normalized cuts.

This requires a modification of the notion of normalized cut.

This new notion it is quite reasonable, as we will see shortly.

If (V, W) is a signed graph, where W is an $m \times m$ symmetric matrix with zero diagonal entries and with the other entries $w_{ij} \in \mathbb{R}$ arbitrary, for any node $v_i \in V$, the *signed degree* of v_i is defined as

$$\bar{d}_i = \bar{d}(v_i) = \sum_{j=1}^m |w_{ij}|,$$

and the *signed degree matrix* $\bar{D}$ as

$$\bar{D} = \text{diag}(\bar{d}(v_1), \dots, \bar{d}(v_m)).$$

For any subset A of the set of nodes V , let

$$\text{vol}(A) = \sum_{v_i \in A} \bar{d}_i = \sum_{v_i \in A} \sum_{j=1}^m |w_{ij}|.$$

For any two subsets A and B of V , define $\text{links}^+(A, B)$, $\text{links}^-(A, B)$, and $\text{cut}(A, \bar{A})$ by

$$\begin{aligned}\text{links}^+(A, B) &= \sum_{\substack{v_i \in A, v_j \in B \\ w_{ij} > 0}} w_{ij} \\ \text{links}^-(A, B) &= \sum_{\substack{v_i \in A, v_j \in B \\ w_{ij} < 0}} -w_{ij} \\ \text{cut}(A, \bar{A}) &= \sum_{\substack{v_i \in A, v_j \in \bar{A} \\ w_{ij} \neq 0}} |w_{ij}|.\end{aligned}$$

Note that $\text{links}^+(A, B) = \text{links}^+(B, A)$, $\text{links}^-(A, B) = \text{links}^-(B, A)$, and

$$\text{cut}(A, \bar{A}) = \text{links}^+(A, \bar{A}) + \text{links}^-(A, \bar{A}).$$

Then, the *signed Laplacian* $\overline{L}$ is defined by

$$\overline{L} = \overline{D} - W,$$

and its normalized version $\overline{L}_{\text{sym}}$ by

$$\overline{L}_{\text{sym}} = \overline{D}^{-1/2} \overline{L} \overline{D}^{-1/2} = I - \overline{D}^{-1/2} W \overline{D}^{-1/2}.$$

For a graph without isolated vertices, we have $\overline{d}(v_i) > 0$ for $i = 1, \dots, m$, so $\overline{D}^{-1/2}$ is well defined.

The signed Laplacian is symmetric positive semidefinite. As for the Laplacian of a weight matrix (with nonnegative entries), this can be shown in two ways.

The first method consists in defining a notion of incidence matrix for a signed graph, and appears in Hou [10].

Definition 4.1. Given a signed graph $G = (V, W)$, with $V = \{v_1, \dots, v_m\}$, if $\{e_1, \dots, e_n\}$ are the edges of the underlying graph of G (recall that $\{v_i, v_j\}$ is an edge of this graph iff $w_{ij} \neq 0$), for any oriented graph G^σ obtained by giving an orientation to the underlying graph of G , the *incidence matrix* B^σ of G^σ is the $m \times n$ matrix whose entries b_{ij} are given by

$$b_{ij} = \begin{cases} +\sqrt{w_{ij}} & \text{if } w_{ij} > 0 \text{ and } s(e_j) = v_i \\ -\sqrt{w_{ij}} & \text{if } w_{ij} > 0 \text{ and } t(e_j) = v_i \\ \sqrt{-w_{ij}} & \text{if } w_{ij} < 0 \text{ and } (s(e_j) = v_i \text{ or } t(e_j) = v_i) \\ 0 & \text{otherwise.} \end{cases}$$

Then, we have the following proposition whose proof is easily adapted from the proof of Proposition 1.2.

Proposition 4.1. *Given any signed graph $G = (V, W)$ with $V = \{v_1, \dots, v_m\}$, if B^σ is the incidence matrix of any oriented graph G^σ obtained from the underlying graph of G and $\overline{D}$ is the signed degree matrix of W , then*

$$B^\sigma (B^\sigma)^\top = \overline{D} - W = \overline{L}.$$

Consequently, $B^\sigma (B^\sigma)^\top$ is independent of the orientation of the underlying graph of G and $\overline{L} = \overline{D} - W$ is symmetric and positive semidefinite; that is, the eigenvalues of $\overline{L} = \overline{D} - W$ are real and nonnegative.

Another way to prove that $\overline{L}$ is positive semidefinite is to evaluate the quadratic form $x^\top \overline{L} x$.

We will need this computation to figure out what is the new notion of normalized cut.

For any real $\lambda \in \mathbb{R}$, define $\text{sgn}(\lambda)$ by

$$\text{sgn}(\lambda) = \begin{cases} +1 & \text{if } \lambda > 0 \\ -1 & \text{if } \lambda < 0 \\ 0 & \text{if } \lambda = 0. \end{cases}$$

Proposition 4.2. *For any $m \times m$ symmetric matrix $W = (w_{ij})$, if we let $\bar{L} = \bar{D} - W$ where $\bar{D}$ is the signed degree matrix associated with W , then we have*

$$x^\top \bar{L} x = \frac{1}{2} \sum_{i,j=1}^m |w_{ij}| (x_i - \text{sgn}(w_{ij}) x_j)^2 \quad \text{for all } x \in \mathbb{R}^m.$$

Consequently, $\bar{L}$ is positive semidefinite.

4.2 Signed Normalized Cuts

As in Section 3.3, given a partition of V into K clusters $(A_1, \dots, A_K)$, if we represent the j th block of this partition by a vector X^j such that

$$X_i^j = \begin{cases} a_j & \text{if } v_i \in A_j \\ 0 & \text{if } v_i \notin A_j, \end{cases}$$

for some $a_j \neq 0$, then we have the following result.

Proposition 4.3. *For any vector X^j representing the j th block of a partition $(A_1, \dots, A_K)$ of V , we have*

$$(X^j)^\top \overline{L} X^j = a_j^2 (\text{cut}(A_j, \overline{A_j}) + 2\text{links}^-(A_j, A_j)).$$

Since with the revised definition of $\text{vol}(A_j)$, we also have

$$(X^j)^\top \overline{D} X^j = a_j^2 \sum_{v_i \in A_j} \overline{d}_i = a_j^2 \text{vol}(A_j),$$

we deduce that

$$\frac{(X^j)^\top \overline{L} X^j}{(X^j)^\top \overline{D} X^j} = \frac{\text{cut}(A_j, \overline{A_j}) + 2\text{links}^-(A_j, A_j)}{\text{vol}(A_j)}.$$

The calculations of the previous paragraph suggest the following definition.

Definition 4.2. The *signed normalized cut* $\text{sNcut}(A_1, \dots, A_K)$ of the partition $(A_1, \dots, A_K)$ is defined as

$$\begin{aligned} \text{sNcut}(A_1, \dots, A_K) = & \sum_{j=1}^K \frac{\text{cut}(A_j, \overline{A_j})}{\text{vol}(A_j)} \\ & + 2 \sum_{j=1}^K \frac{\text{links}^-(A_j, A_j)}{\text{vol}(A_j)}. \end{aligned}$$

Remark: Kunegis et al. [12] deal with a different notion of cut, namely ratio cut (in which $\text{vol}(A)$ is replaced by the size $|A|$ of A), and only for two clusters.

In this case, by a clever choice of indicator vector, they obtain a notion of signed cut that only takes into account the positive edges between A and $\overline{A}$, and the negative edges among nodes in A and nodes in $\overline{A}$.

This trick does not seem to generalize to more than two clusters, and this is why we use our representation for partitions. Our definition of a signed normalized cut appears to be novel.

Based on previous computations, we have

$$\text{sNcut}(A_1, \dots, A_K) = \sum_{j=1}^K \frac{(X^j)^\top \overline{L} X^j}{(X^j)^\top \overline{D} X^j}.$$

where X is the $N \times K$ matrix whose j th column is X^j .

Therefore, this is the same problem as in Chapter 3, with L replaced by $\overline{L}$ and D replaced by $\overline{D}$.

Observe that minimizing $\text{sNcut}(A_1, \dots, A_K)$ amounts to minimizing the number of positive and negative edges between clusters, and also minimizing the number of negative edges within clusters.

This second minimization captures the intuition that *nodes connected by a negative edge should not be together* (they do not “like” each other; they should be far from each other).

The K -clustering problem for signed graphs is related but not equivalent to another problem known as *correlation clustering*.

In correlation clustering, in our terminology and notation, given a graph $G = (V, W)$ with positively and negatively weighted edges, one seeks a clustering of V that minimizes the sum $\text{links}^-(A_j, A_j)$ of the absolute values of the negative weights of the edges within each cluster A_j , and minimizes the sum $\text{links}^+(A_j, \bar{A}_j)$ of the positive weights of the edges between distinct clusters.

In contrast to K -clustering, the number K of clusters is not given in advance, and there is no normalization with respect to size of volume.

Furthermore, in correlation clustering, only the contribution $\text{links}^+(A_j, \overline{A}_j)$ of positively weighted edges is minimized, but our method only allows us to minimize $\text{cut}(A_j, \overline{A}_j)$, which also takes into account negatively weighted edges between distinct clusters.

Correlation clustering was first introduced and studied for complete graphs by Bansal, Blum and Chawla [1].

They prove that this problem is NP-complete and give several approximation algorithms, including a PTAS for maximizing agreement.

Demaine and Immorlica [4] consider the same problem for arbitrary weighted graphs, and they give an $O(\log n)$ -approximation algorithm based on linear programming.

Since correlation clustering does not assume that K is given and not include normalization by size or volume, it is not clear whether algorithms for correlation clustering can be applied to normalized K -clustering, and conversely.

4.3 Balanced Graphs

Since

$$\text{sNcut}(A_1, \dots, A_K) = \sum_{j=1}^K \frac{(X^j)^\top \overline{L} X^j}{(X^j)^\top \overline{D} X^j},$$

the whole machinery of Sections 3.3 and 3.4 can be applied with D replaced by $\overline{D}$ and L replaced by $\overline{L}$.

However, there is a new phenomenon, which is that $\overline{L}$ *blue may be positive definite*.

As a consequence, $\mathbf{1}$ is not always an eigenvector of $\overline{L}$.

As observed by Kunegis et al. [12], it is also possible to characterize for which signed graphs the Laplacian $\overline{L}$ is positive definite.

Such graphs are “cousins” of bipartite graphs and were introduced by Harary [9].

Since a graph is the union of its connected components, we restrict ourselves to connected graphs.

Definition 4.3. Given a signed graph $G = (V, W)$ with negative weights whose underlying graph is connected, we say that G is *balanced* if there is a partition of its set of nodes V into two blocks V_1 and V_2 such that all positive edges connect nodes within V_1 or V_2 , and negative edges connect nodes between V_1 and V_2 .

An example of a balanced graph is shown in Figure 4.1 on the left, in which positive edges are colored green and negative edges are colored red.

This graph admits the partition

$$(\{v_1, v_2, v_4, v_7, v_8\}, \{v_3, v_5, v_6, v_9\}).$$

On the other hand, the graph shown on the right contains the cycle $(v_2, v_3, v_6, v_5, v_4, v_2)$ with an odd number of negative edges (3), and thus is not balanced.

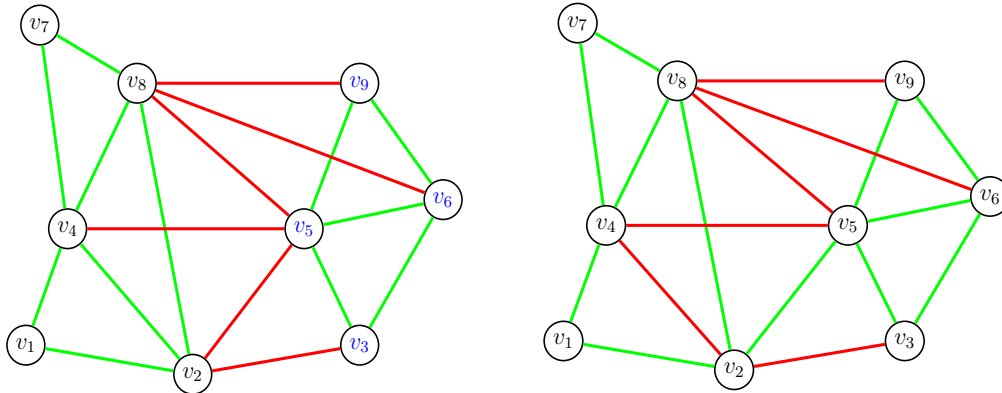


Figure 4.1: A balanced signed graph G_1 (left). An unbalanced signed graph G_2 (right).

Observe that if we delete all positive edges in a balanced graph, then the resulting graph is bipartite.

Then, it is not surprising that connected balanced graphs can be characterized as signed graphs in which every cycle has an even number of negative edges.

This is analogous to the characterization of a connected bipartite graph as a graph in which every cycle has even length.

The following proposition was first proved by Harary [9].

Proposition 4.4. *If $G = (V, W)$ is a connected signed graph with negative weights, then G is balanced iff every cycle contains an even number of negative edges.*

We can also detect whether a connected signed graph is balanced in terms of the kernel of the transpose of any of its incidence matrices.

Proposition 4.5. *If $G = (V, W)$ is a connected signed graph with negative weights and with m nodes, for any orientation of its underlying graph, let B be the corresponding incidence matrix. The underlying graph of G is balanced iff $\text{rank}(B) = m - 1$. Furthermore, if G is balanced, then there is a vector u with $u_i \in \{-1, 1\}$ such that $B^\top u = 0$, and the sets of nodes $V_1 = \{v_i \mid u_i = -1\}$ and $V_2 = \{v_i \mid u_i = +1\}$ form a partition of V for which G is balanced.*

Remark: A simple modification of the proof of Proposition 4.5 shows that if there are c_1 components containing only positive edges, c_2 components that are balanced graphs, and c_3 components that are not balanced (and contain some negative edge), then

$$c_1 + c_2 = m - \text{rank}(B).$$

Since by Proposition 4.1 we have $\overline{L} = BB^\top$ for any incidence matrix B associated with an orientation of the underlying graph of G , we obtain the following important result (which is proved differently in Kunegis et al. [12]).

Theorem 4.6. *The signed Laplacian $\overline{L}$ of a connected signed graph G is positive definite iff G is not balanced (possesses some cycle with an odd number of negative edges).*

If $G = (V, W)$ is a balanced graph, then there is a partition (V_1, V_2) of V such that for every edge $\{v_i, v_j\}$, if $w_{ij} > 0$, then $v_i, v_j \in V_1$ or $v_i, v_j \in V_2$, and if $w_{ij} < 0$, then $v_i \in V_1$ and $v_j \in V_2$.

It follows that if we define the vector x such that $x_i = +1$ iff $v_i \in V_1$ and $x_i = -1$ iff $v_i \in V_2$, then for every edge $\{v_i, v_j\}$ we have

$$\text{sgn}(w_{ij}) = x_i x_j.$$

We call x a *bipartition* of V .

The signed Laplacian of the balanced graph G_1 is given by

$$\overline{L}_1 = \begin{pmatrix} 2 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 5 & 1 & -1 & 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 3 & 0 & -1 & -1 & 0 & 0 & 0 \\ -1 & -1 & 0 & 5 & 1 & 0 & -1 & -1 & 0 \\ 0 & 1 & -1 & 1 & 6 & -1 & 0 & 1 & -1 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 1 & -1 \\ 0 & 0 & 0 & -1 & 0 & 0 & 2 & -1 & 0 \\ 0 & -1 & 0 & -1 & 1 & 1 & -1 & 6 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & 1 & 3 \end{pmatrix}$$

Using **Matlab**, we find that its eigenvalues are

$$0, 1.4790, 1.7513, 2.7883, 4.3570, 4.8815, 6.2158, \\ 7.2159, 7.3112.$$

The eigenvector corresponding to the eigenvalue 0 is

$$(0.3333, 0.3333, -0.3333, 0.3333, -0.3333, \\ -0.3333, 0.3333, 0.3333, -0.3333)$$

It gives us the bipartition

$$(\{v_1, v_2, v_4, v_7, v_8\}, \{v_3, v_5, v_6, v_9\}),$$

as guaranteed by Proposition 4.5.

The signed Laplacian of the unbalanced graph G_2 is given by

$$\overline{L}_2 = \begin{pmatrix} 2 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 5 & 1 & 1 & -1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 3 & 0 & -1 & -1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 5 & 1 & 0 & -1 & -1 & 0 \\ 0 & -1 & -1 & 1 & 6 & -1 & 0 & 1 & -1 \\ 0 & 0 & -1 & 0 & -1 & 4 & 0 & 1 & -1 \\ 0 & 0 & 0 & -1 & 0 & 0 & 2 & -1 & 0 \\ 0 & -1 & 0 & -1 & 1 & 1 & -1 & 6 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & 1 & 3 \end{pmatrix}$$

The eigenvalues of $\overline{L}_2$ are

$$0.5175, 1.5016, 1.7029, 2.7058, 3.7284, 4.9604, 5.6026, \\ 7.0888, 8.1921.$$

The matrix $\overline{L}_2$ is indeed positive definite (since G_2 is unbalanced).

Hou [10] gives bounds on the smallest eigenvalue of an unbalanced graph. The lower bound involves a measure of how unbalanced the graph is (see Theorem 3.4 in Hou [10]).

Following Kunegis et al., we can prove the following result showing that the eigenvalues and the eigenvectors of $\overline{L}$ and its unsigned counterpart $\mathcal{L}$ are strongly related.

Given a symmetric signed matrix W , we define the unsigned matrix $\mathcal{W}$ such that $\mathcal{W}_{ij} = |w_{ij}|$ ($1 \leq i, j \leq m$). We let $\mathcal{L}$ be the Laplacian associated with $\mathcal{W}$.

Note that

$$\mathcal{L} = \overline{D} - \mathcal{W}.$$

The following proposition is shown in Kunegis et al. [12]).

Proposition 4.7. *Let $G = (V, W)$ be a signed graph and let $\mathcal{W}$ be the unsigned matrix associated with W . If G is balanced, and x is a bipartition of V , then for any diagonalization $\bar{L} = P\Lambda P^\top$ of $\bar{L}$, where P is an orthogonal matrix of eigenvectors of $\bar{L}$, if we define the matrix $\mathcal{P}$ so that*

$$\mathcal{P}_i = x_i P_i,$$

where $\mathcal{P}_i$ is the i th row of $\mathcal{P}$ and P_i is the i th row of P , then $\mathcal{P}$ is orthogonal and

$$\mathcal{L} = \mathcal{P}\Lambda\mathcal{P}^\top$$

is a diagonalization of $\mathcal{L}$. In particular, $\bar{L}$ and $\mathcal{L}$ have the same eigenvalues with the same multiplicities.

4.4 *K*-Way Clustering of Signed Graphs

Using the signed Laplacians $\bar{L}$ and $\bar{L}_{\text{sym}}$, we can define the optimization problems as in Section 3.3 and solve them as in Section 3.4, except that we drop the constraint

$$X(X^\top X)^{-1}X^\top \mathbf{1} = \mathbf{1},$$

since $\mathbf{1}$ is not necessarily an eigenvector of $\bar{L}$.

By Proposition A.3, the sum of the K smallest eigenvalues of $\bar{L}_{\text{sym}}$ is a lower bound for $\text{tr}(Y^\top \bar{L}_{\text{sym}} Y)$, and the minimum of problem $(**_2)$ is achieved by any K unit eigenvectors $(u_1, \dots, u_k)$ associated with the smallest eigenvalues

$$0 \leq \nu_1 \leq \nu_2 \leq \dots \leq \nu_K$$

of $\bar{L}_{\text{sym}}$.

The difference with unsigned graphs is that ν_1 may be strictly positive.

Here is the result of applying this method to various examples.

First, we apply our algorithm to find three clusters for the balanced graph G_1 .

The graph G_1 as outputted by the algorithm is shown in Figure 4.2 and the three clusters are shown in Figure 4.3.

As desired, these clusters do not contain negative edges.

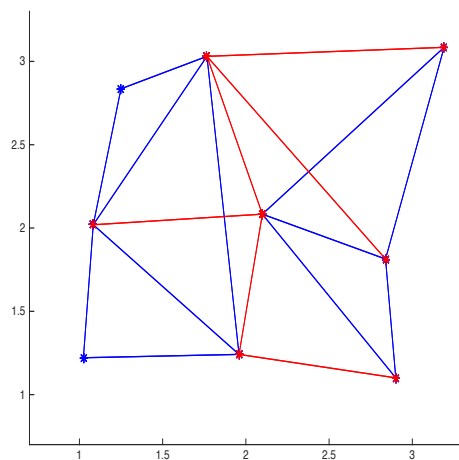


Figure 4.2: The balanced graph G_1 .

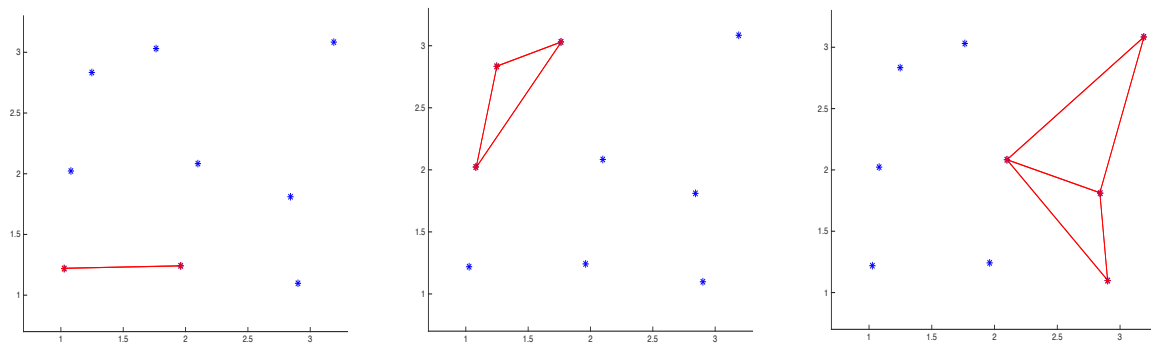
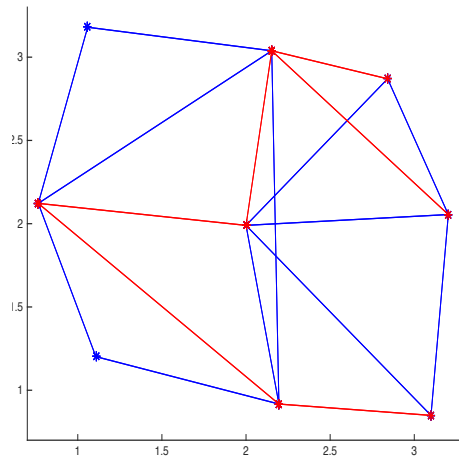
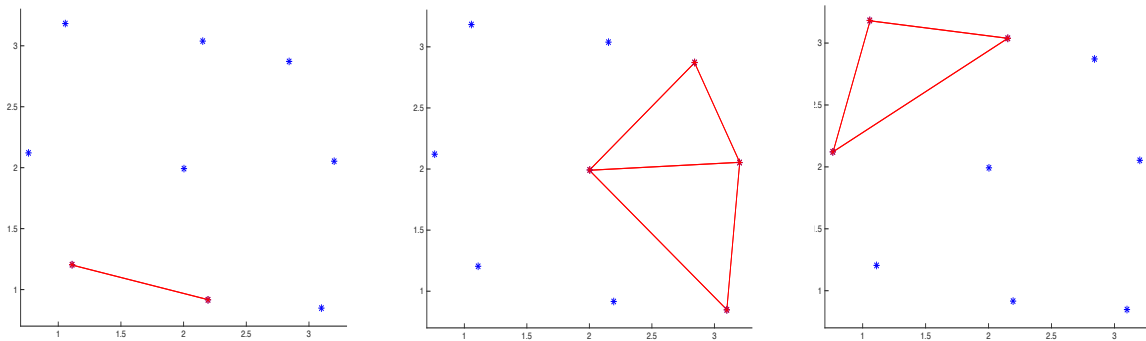


Figure 4.3: Three blocks of a normalized cut for the graph associated with G_1 .

By the way, for two clusters, the algorithm finds the bipartition of G_1 , as desired.

Next, we apply our algorithm to find three clusters for the unbalanced graph G_2 . The graph G_2 as outputted by the algorithm is shown in Figure 4.2 and the three clusters are shown in Figure 4.3.

As desired, these clusters do not contain negative edges.

Figure 4.4: The unbalanced graph G_2 .Figure 4.5: Three blocks of a normalized cut for the graph associated with G_2 .

The algorithm finds the same clusters, but this is probably due to the fact that G_1 and G_2 only differ by the signs of two edges.

4.5 Signed Graph Drawing

Following Kunegis et al. [12], if our goal is to draw a signed graph $G = (V, W)$ with m nodes, a natural way to interpret negative weights is to assume that the endpoints v_i and v_j of an edge with a negative weight should be placed far apart, which can be achieved if instead of assigning the point $\rho(v_j) \in \mathbb{R}^n$ to v_j , we assign the point $-\rho(v_j)$.

Then, if R is the $m \times n$ matrix of a graph drawing of G in $\mathbb{R}^n$, the energy function $\mathcal{E}(R)$ is redefined to be

$$\mathcal{E}(R) = \sum_{\{v_i, v_j\} \in E} |w_{ij}| \|\rho(v_i) - \text{sgn}(w_{ij})\rho(v_j)\|^2.$$

We obtain the following version of Proposition 2.1.

Proposition 4.8. *Let $G = (V, W)$ be a signed graph, with $|V| = m$ and with W a $m \times m$ symmetric matrix, and let R be the matrix of a graph drawing ρ of G in $\mathbb{R}^n$ (a $m \times n$ matrix). Then, we have*

$$\mathcal{E}(R) = \text{tr}(R^\top \overline{L} R).$$

Then, as in Chapter 2, we look for a graph drawing R that minimizes $\mathcal{E}(R) = \text{tr}(R^\top \overline{L} R)$ subject to $R^\top R = I$.

The new ingredient is that $\overline{L}$ is positive definite iff G is not a balanced graph.

Also, in the case of a signed graph, $\mathbf{1}$ does not belong to the kernel of $\overline{L}$, so we do not get a balanced graph drawing.

If G is a signed balanced graph, then $\text{Ker } L$ is nontrivial, and if G is connected, then $\text{Ker } L$ is spanned by a vector whose components are either $+1$ or -1 .

Thus, if we use the first n unit eigenvectors $(u_1, u_2, \dots, u_n)$ associated with the n smallest eigenvalues $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_n$ of $\overline{L}$, we obtain a drawing for which the nodes are partitioned into two sets living in two hyperplanes corresponding to the value of their first coordinate.

Let us call such a drawing a *bipartite drawing*.

However, if G is connected, the vector u_2 does not belong to $\text{Ker } \overline{L}$, so if $m \geq 3$, it must have at least three coordinates with distinct absolute values, and using $(u_2, \dots, u_{n+1})$ we obtain a nonbipartite graph.

Then, the following version of Theorem 2.2 is easily shown.

Theorem 4.9. *Let $G = (V, W)$ be a signed graph with $|V| = m \geq 3$, assume that G has some negative edge and is connected, and let $\bar{L} = \bar{D} - W$ be the signed Laplacian of G .*

- (1) *If G is not balanced and if the eigenvalues of L are $0 < \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_m$, then the minimal energy of any orthogonal graph drawing of G in $\mathbb{R}^n$ is equal to $\lambda_1 + \dots + \lambda_n$. The $m \times n$ matrix R consisting of any unit eigenvectors $u_1, \dots, u_n$ associated with $\lambda_1 \leq \dots \leq \lambda_n$ yields an orthogonal graph drawing of minimal energy.*
- (2) *If G is balanced and if the eigenvalues of L are $0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_m$, then the minimal energy of any orthogonal nonbipartite graph drawing of G in $\mathbb{R}^n$ is equal to $\lambda_2 + \dots + \lambda_{n+1}$ (in particular, this implies that $n < m$). The $m \times n$ matrix R consisting of any unit eigenvectors $u_2, \dots, u_{n+1}$ associated with $\lambda_2 \leq \dots \leq \lambda_{n+1}$ yields an orthogonal nonbipartite graph drawing of minimal energy.*

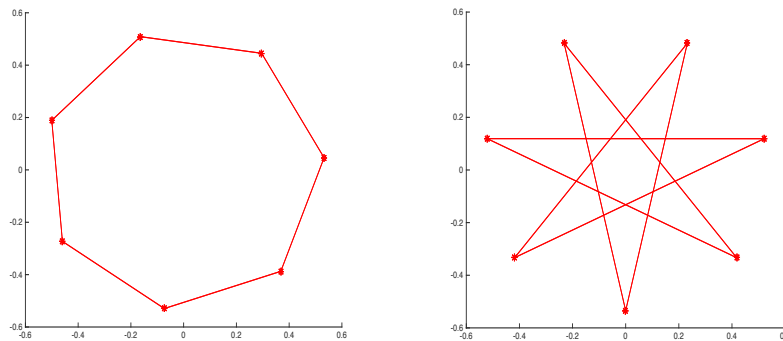
(3) *If G is balanced, for $n = 2$, a graph drawing of G as a bipartite graph (with positive edges only within the two blocks of vertices) is obtained from the $m \times 2$ matrix consisting of any two unit eigenvectors u_1 and u_2 associated with 0 and λ_2 .*

In all cases, the graph drawing R satisfies the condition $R^\top R = I$ (it is an orthogonal graph drawing).

Our first example is the signed graph G_4 defined by the weight matrix given by the following **Matlab** program:

```
nn = 6; G3 = diag(ones(1,nn),1);
G3 = G3 + G3';
G3(1,nn+1) = 1; G3(nn+1,1) = 1;
G4 = -G3;
```

All edges of this graph are negative. The graph obtained by using G_3 is shown on the left and the graph obtained by using the signed Laplacian of G_4 is shown on the right in Figure 4.6.

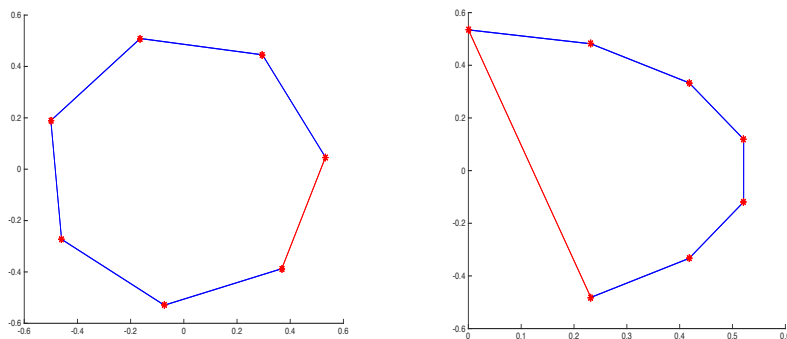
Figure 4.6: The signed graph $G4$.

The second example is the signed graph $G5$ obtained from $G3$ by making a single edge negative:

$$G5 = G3; \quad G5(1,2) = -1; \quad G5(2,1) = -1;$$

The graph obtained by using $G3$ is shown on the left and the graph obtained by using the signed Laplacian of $G5$ is shown on the right in Figure 4.7.

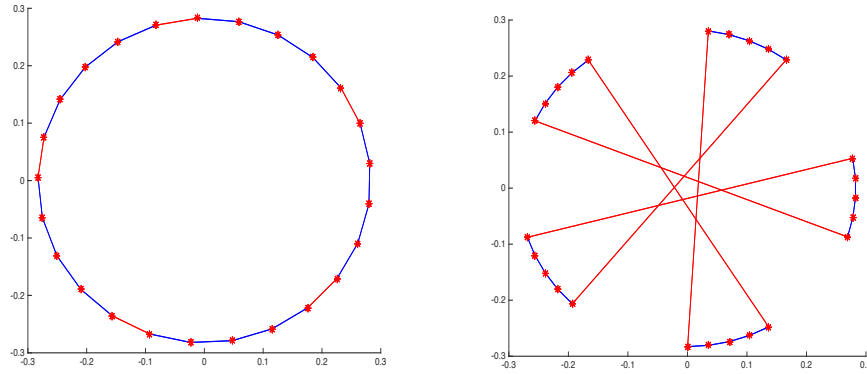
Positive edges are shown in blue and negative edges are shown in red.

Figure 4.7: The signed graph $G5$.

The third example is the signed graph $G6$ defined by the weight matrix given by the following **Matlab** program:

```
nn = 24; G6 = diag(ones(1,nn),1);
G6 = G6 + G6';
G6(1,nn+1) = 1; G6(nn+1,1) = 1;
G6(1,2) = -1; G6(2,1) = -1; G6(6,7) = -1;
G6(7,6) = -1; G6(11,12) = -1; G6(12,11) = -1;
G6(16,17) = -1; G6(17,16) = -1;
G6(21,22) = -1; G6(22,21) = -1;
```

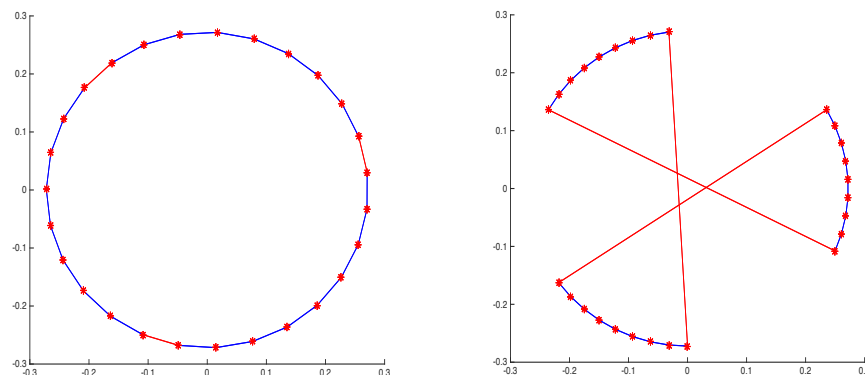
The graph obtained by using absolute values in $G6$ is shown on the left and the graph obtained by using the signed Laplacian of $G6$ is shown on the right in Figure 4.8.

Figure 4.8: The signed graph $G6$.

The fourth example is the signed graph $G7$ defined by the weight matrix given by the following **Matlab** program:

```
nn = 26; G7 = diag(ones(1,nn),1);
G7 = G7 + G7'; G7(1,nn+1) = 1;
G7(nn+1,1) = 1; G7(1,2) = -1;
G7(2,1) = -1; G7(10,11) = -1;
G7(11,10) = -1; G7(19,20) = -1;
G7(20,19) = -1;
```

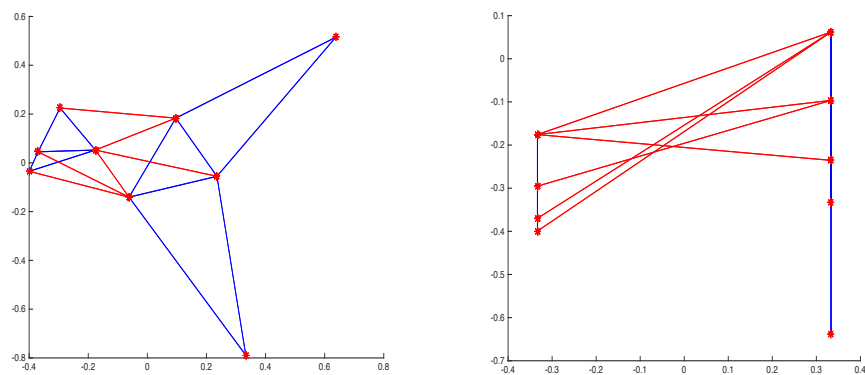
The graph obtained by using absolute values in $G7$ is shown on the left and the graph obtained by using the signed Laplacian of $G7$ is shown on the right in Figure 4.9.

Figure 4.9: The signed graph $G7$.

These graphs are all unbalanced. As predicted, nodes linked by negative edges are far from each other.

Our last example is the balanced graph $G1$ from Figure 4.1.

The graph obtained by using absolute values in $G1$ is shown on the left and the bipartite graph obtained by using the signed Laplacian of $G1$ is shown on the right in Figure 4.10.

Figure 4.10: The balanced graph $G1$.

Chapter 5

Graph Clustering Using Ratio Cuts

In this short chapter, we consider the alternative to normalized cut, *called ratio cut*, and show that the methods of Chapters 3 and 4 can be trivially adapted to solve the clustering problem using ratio cuts.

All that needs to be done is to replace the normalized Laplacian L_{sym} by the unnormalized Laplacian L , and omit the step of considering Problem (**₂).

In particular, there is no need to multiply the continuous solution Y by $D^{-1/2}$.

The idea of ratio cut is to *replace the volume $\text{vol}(A_j)$ of each block A_j of the partition by its size $|A_j|$* (the number of nodes in A_j).

First, we deal with unsigned graphs, the case where the entries in the symmetric weight matrix W are nonnegative.

Definition 5.1. The *ratio cut* $\text{Rcut}(A_1, \dots, A_K)$ of the partition $(A_1, \dots, A_K)$ is defined as

$$\text{Rcut}(A_1, \dots, A_K) = \sum_{j=1}^K \frac{\text{cut}(A_j, \overline{A_j})}{|A_j|}.$$

As in Section 3.3, given a partition of V into K clusters $(A_1, \dots, A_K)$, if we represent the j th block of this partition by a vector X^j such that

$$X_i^j = \begin{cases} a_j & \text{if } v_i \in A_j \\ 0 & \text{if } v_i \notin A_j, \end{cases}$$

for some $a_j \neq 0$, then

$$\begin{aligned} (X^j)^\top L X^j &= a_j^2 (\text{cut}(A_j, \overline{A_j})) \\ (X^j)^\top X^j &= a_j^2 |A_j|. \end{aligned}$$

Consequently, we have

$$\begin{aligned} \text{Rcut}(A_1, \dots, A_K) &= \sum_{j=1}^K \frac{\text{cut}(A_j, \overline{A_j})}{|A_j|} \\ &= \sum_{j=1}^K \frac{(X^j)^\top L X^j}{(X^j)^\top X^j}. \end{aligned}$$

On the other hand, the normalized cut is given by

$$\begin{aligned} \text{Ncut}(A_1, \dots, A_K) &= \sum_{j=1}^K \frac{\text{cut}(A_j, \overline{A_j})}{\text{vol}(A_j)} \\ &= \sum_{j=1}^K \frac{(X^j)^\top L X^j}{(X^j)^\top D X^j}. \end{aligned}$$

Therefore, ratio cut is the special case of normalized cut where $D = I$.

If we let

$$\mathcal{X} = \left\{ [X^1 \dots X^K] \mid X^j = a_j(x_1^j, \dots, x_N^j), x_i^j \in \{1, 0\}, \right. \\ \left. a_j \in \mathbb{R}, X^j \neq 0 \right\}$$

(note that the condition $X^j \neq 0$ implies that $a_j \neq 0$), then the set of matrices representing partitions of V into K blocks is

$$\mathcal{K} = \left\{ X = [X^1 \ \dots \ X^K] \mid \begin{aligned} &X \in \mathcal{X}, \\ &(X^i)^\top X^j = 0, \\ &1 \leq i, j \leq K, \ i \neq j \end{aligned} \right\}.$$

Here is our first formulation of K -way clustering of a graph using ratio cuts, called problem PRC1 :

**K -way Clustering of a graph using Ratio Cut,
Version 1:
Problem PRC1**

$$\begin{aligned} &\text{minimize} && \sum_{j=1}^K \frac{(X^j)^\top L X^j}{(X^j)^\top X^j} \\ &\text{subject to} && (X^i)^\top X^j = 0, \quad 1 \leq i, j \leq K, \ i \neq j, \\ &&& X \in \mathcal{X}. \end{aligned}$$

The solutions that we are seeking are K -tuples $(\mathbb{P}(X^1), \dots, \mathbb{P}(X^K))$ of points in $\mathbb{RP}^{N-1}$ determined by their homogeneous coordinates $X^1, \dots, X^K$.

As in Chapter 3, chasing denominators and introducing a trace, we obtain the following formulation of our minimization problem:

**K -way Clustering of a graph using Ratio Cut,
Version 2:
Problem PRC2**

$$\begin{array}{ll} \text{minimize} & \text{tr}(X^\top L X) \\ \text{subject to} & X^\top X = I, \\ & X \in \mathcal{X}. \end{array}$$

The natural relaxation of problem PRC2 is to drop the condition that $X \in \mathcal{X}$, and we obtain the

Problem ($R*_2$)

$$\begin{array}{ll} \text{minimize} & \text{tr}(X^\top L X) \\ \text{subject to} & X^\top X = I. \end{array}$$

This time, since the normalization condition is $X^\top X = I$, we can use the eigenvalues and the eigenvectors of L , and by Proposition A.2, the minimum is achieved by any K unit eigenvectors $(u_1, \dots, u_K)$ associated with the smallest K eigenvalues

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_K$$

of L .

The matrix $Z = Y = [u_1, \dots, u_K]$ yields a minimum of our relaxed problem $(R*_2)$.

The rest of the algorithm is as before; we try to find $Q = R\Lambda$ with $R \in \mathbf{O}(K)$, Λ diagonal invertible, and $X \in \mathcal{X}$ such that $\|X - ZQ\|$ is minimum.

In the case of signed graphs, we define the *signed ratio cut* $\text{sRcut}(A_1, \dots, A_K)$ of the partition $(A_1, \dots, A_K)$ as

$$\begin{aligned} \text{sRcut}(A_1, \dots, A_K) = \sum_{j=1}^K \frac{\text{cut}(A_j, \overline{A_j})}{|A_j|} \\ + 2 \sum_{j=1}^K \frac{\text{links}^-(A_j, A_j)}{|A_j|}. \end{aligned}$$

Since we still have

$$(X^j)^\top \overline{L} X^j = a_j^2 (\text{cut}(A_j, \overline{A_j}) + 2\text{links}^-(A_j, A_j)),$$

we obtain

$$\text{sRcut}(A_1, \dots, A_K) = \sum_{j=1}^K \frac{(X^j)^\top \overline{L} X^j}{(X^j)^\top X^j}.$$

Therefore, this is similar to the case of unsigned graphs, with L replaced with $\overline{L}$.

The same algorithm applies, but as in Chapter 4, the signed Laplacian $\overline{L}$ is positive definite iff G is unbalanced.

Modifying the computer program implementing normalized cuts to deal with ratio cuts is trivial (use $\bar{L}$ instead of $\bar{L}_{\text{sym}}$ and don't multiply Y by $\bar{D}^{-1/2}$).

Generally, normalized cut seems to yield “better clusters,” but this is not a very satisfactory statement since we haven't defined precisely in which sense a clustering is better than another.

We leave this point as further research.

Appendix A

Rayleigh Ratios and the Courant-Fischer Theorem

The most important property of symmetric matrices is that they have real eigenvalues and that they can be diagonalized with respect to an orthogonal matrix.

Thus, if A is an $n \times n$ symmetric matrix, then it has n real eigenvalues $\lambda_1, \dots, \lambda_n$ (not necessarily distinct), and there is an orthonormal basis of eigenvectors $(u_1, \dots, u_n)$ (for a proof, see Gallier [6]).

Another fact that is used frequently in optimization problem is that the eigenvalues of a symmetric matrix are characterized in terms of what is known as the *Rayleigh ratio*, defined by

$$R(A)(x) = \frac{x^\top Ax}{x^\top x}, \quad x \in \mathbb{R}^n, x \neq 0.$$

The following proposition is often used to prove the correctness of various optimization or approximation problems (for example PCA).

Proposition A.1. (*Rayleigh–Ritz*) If A is a symmetric $n \times n$ matrix with eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and if $(u_1, \dots, u_n)$ is any orthonormal basis of eigenvectors of A , where u_i is a unit eigenvector associated with λ_i , then

$$\max_{x \neq 0} \frac{x^\top A x}{x^\top x} = \lambda_n$$

(with the maximum attained for $x = u_n$), and

$$\max_{x \neq 0, x \in \{u_{n-k+1}, \dots, u_n\}^\perp} \frac{x^\top A x}{x^\top x} = \lambda_{n-k}$$

(with the maximum attained for $x = u_{n-k}$), where $1 \leq k \leq n - 1$.

Equivalently, if V_k is the subspace spanned by $(u_1, \dots, u_k)$, then

$$\lambda_k = \max_{x \neq 0, x \in V_k} \frac{x^\top A x}{x^\top x}, \quad k = 1, \dots, n.$$

For our purposes, we also need the version of Proposition A.1 applying to min instead of max.

Proposition A.2. (*Rayleigh–Ritz*) *If A is a symmetric $n \times n$ matrix with eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ and if $(u_1, \dots, u_n)$ is any orthonormal basis of eigenvectors of A , where u_i is a unit eigenvector associated with λ_i , then*

$$\min_{x \neq 0} \frac{x^\top A x}{x^\top x} = \lambda_1$$

(with the minimum attained for $x = u_1$), and

$$\min_{x \neq 0, x \in \{u_1, \dots, u_{i-1}\}^\perp} \frac{x^\top A x}{x^\top x} = \lambda_i$$

(with the minimum attained for $x = u_i$), where $2 \leq i \leq n$.

Equivalently, if $W_k = V_{k-1}^\perp$ denotes the subspace spanned by $(u_k, \dots, u_n)$ (with $V_0 = (0)$), then

$$\lambda_k = \min_{x \neq 0, x \in W_k} \frac{x^\top A x}{x^\top x} = \min_{x \neq 0, x \in V_{k-1}^\perp} \frac{x^\top A x}{x^\top x}, \quad k = 1, \dots, n.$$

Propositions A.1 and A.2 together are known as the *Rayleigh–Ritz theorem*.

As an application of Propositions A.1 and A.2, we give a proof of a proposition which is the key to the proof of Theorem 2.2.

Given an $n \times n$ symmetric matrix A and an $m \times m$ symmetric B , with $m \leq n$, if $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ are the eigenvalues of A and $\mu_1 \leq \mu_2 \leq \cdots \leq \mu_m$ are the eigenvalues of B , then we say that the *eigenvalues of B interlace the eigenvalues of A* if

$$\lambda_i \leq \mu_i \leq \lambda_{n-m+i}, \quad i = 1, \dots, m.$$

The following proposition is known as the *Poincaré separation theorem*.

Proposition A.3. *Let A be an $n \times n$ symmetric matrix, R be an $n \times m$ matrix such that $R^\top R = I$ (with $m \leq n$), and let $B = R^\top AR$ (an $m \times m$ matrix). The following properties hold:*

- (a) *The eigenvalues of B interlace the eigenvalues of A .*
- (b) *If $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ are the eigenvalues of A and $\mu_1 \leq \mu_2 \leq \cdots \leq \mu_m$ are the eigenvalues of B , and if $\lambda_i = \mu_i$, then there is an eigenvector v of B with eigenvalue μ_i such that Rv is an eigenvector of A with eigenvalue λ_i .*

Observe that Proposition A.3 implies that

$$\lambda_1 + \cdots + \lambda_m \leq \operatorname{tr}(R^\top AR) \leq \lambda_{n-m+1} + \cdots + \lambda_n.$$

The left inequality is used to prove Theorem 2.2.

For the sake of completeness, we also prove the Courant–Fischer characterization of the eigenvalues of a symmetric matrix.

Theorem A.4. (*Courant–Fischer*) *Let A be a symmetric $n \times n$ matrix with eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and let $(u_1, \dots, u_n)$ be any orthonormal basis of eigenvectors of A , where u_i is a unit eigenvector associated with λ_i . If $\mathcal{V}_k$ denotes the set of subspaces of $\mathbb{R}^n$ of dimension k , then*

$$\lambda_k = \max_{W \in \mathcal{V}_{n-k+1}} \min_{x \in W, x \neq 0} \frac{x^\top A x}{x^\top x}$$

$$\lambda_k = \min_{W \in \mathcal{V}_k} \max_{x \in W, x \neq 0} \frac{x^\top A x}{x^\top x}.$$

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